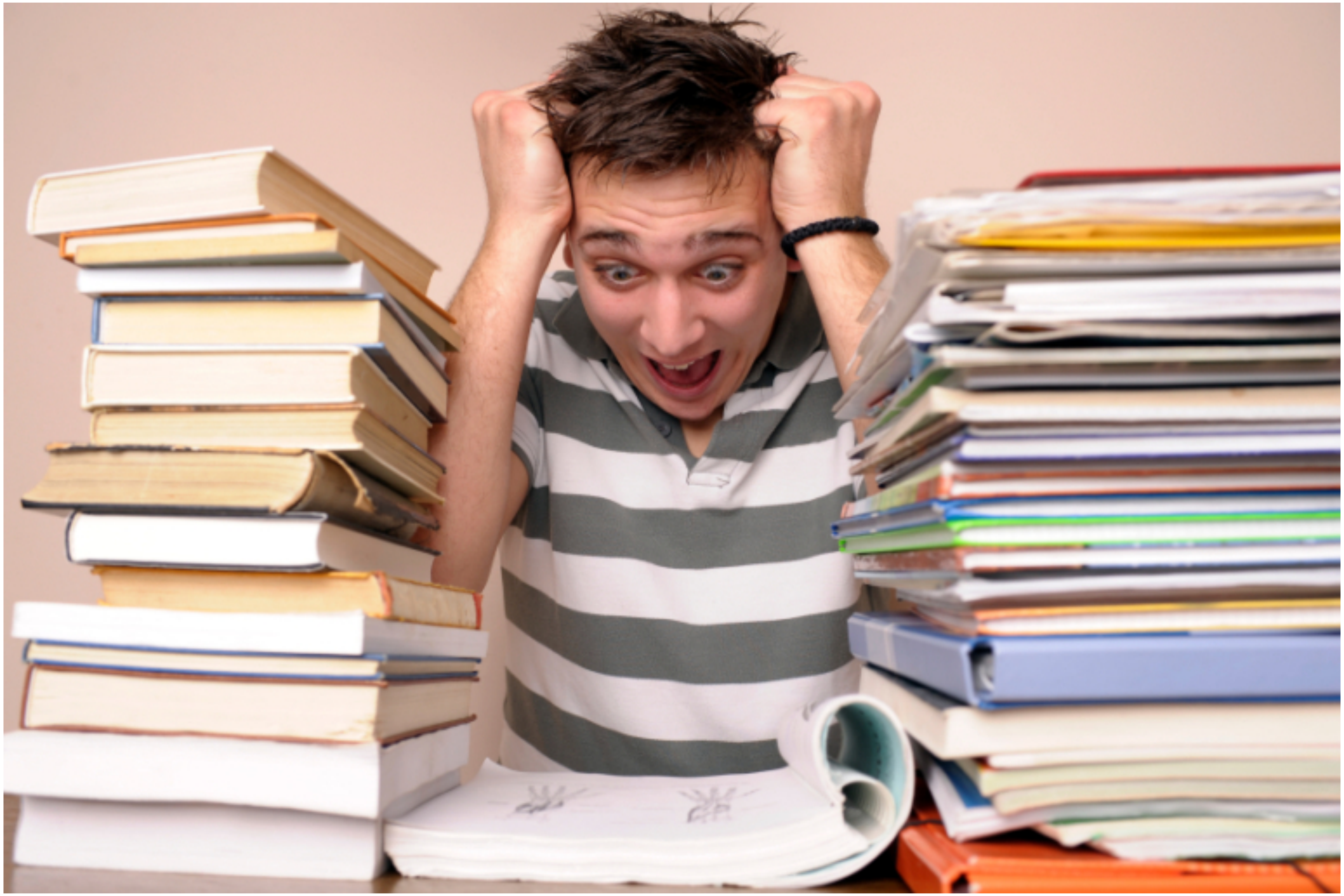
MIDTERM REVIEW



BF527 – Applications in Bioinformatics

On the Blackboard

- **Everyone will have an exam version assigned**

Student -> Exam Version

Username	First	Last	Exam Version
doe	John	Doe	1
jb	Justin	Beeber	2

- On the day of the exam the versions will be posted on the Blackboard

Exam

Attached Files: BF527_Midterm_version10.ipynb (230.391 KB)
BF527_Midterm_version11.ipynb (229.763 KB)
BF527_Midterm_version14.ipynb (229.401 KB)

During the exam

- Exam will be administered in the classroom
- Open book, open Internet, no help from others
- It's OK to use ChatGPT. https://chatgpt.com/ or any other AI tool. Please indicate, which LLM chatbot you use.
- Be succinct
- Partial credit will be given for the partially correct attempts
- No penalty for incorrect answers
- Exactly 1.5 hours from start to finish
- It's OK to ask questions in class, on Slack, or in Zoom chat
  - If you think the class will benefit, ask in a common area, Slack #questions or Zoom chat to everyone

5 multi-part questions on:

1) Sequence alignment
2) BLAST
3) Multiple sequence alignment
4) Phylogenetic trees
5) NextGen and RNA-Seq terms, omics analysis and Pseudocode

Not expected to program on the exam, but programming concepts will be tested (e.g. pseudocode, time complexity).

The exam will be administered in Jupyter notebooks
(make sure you know how to create a new cell!)

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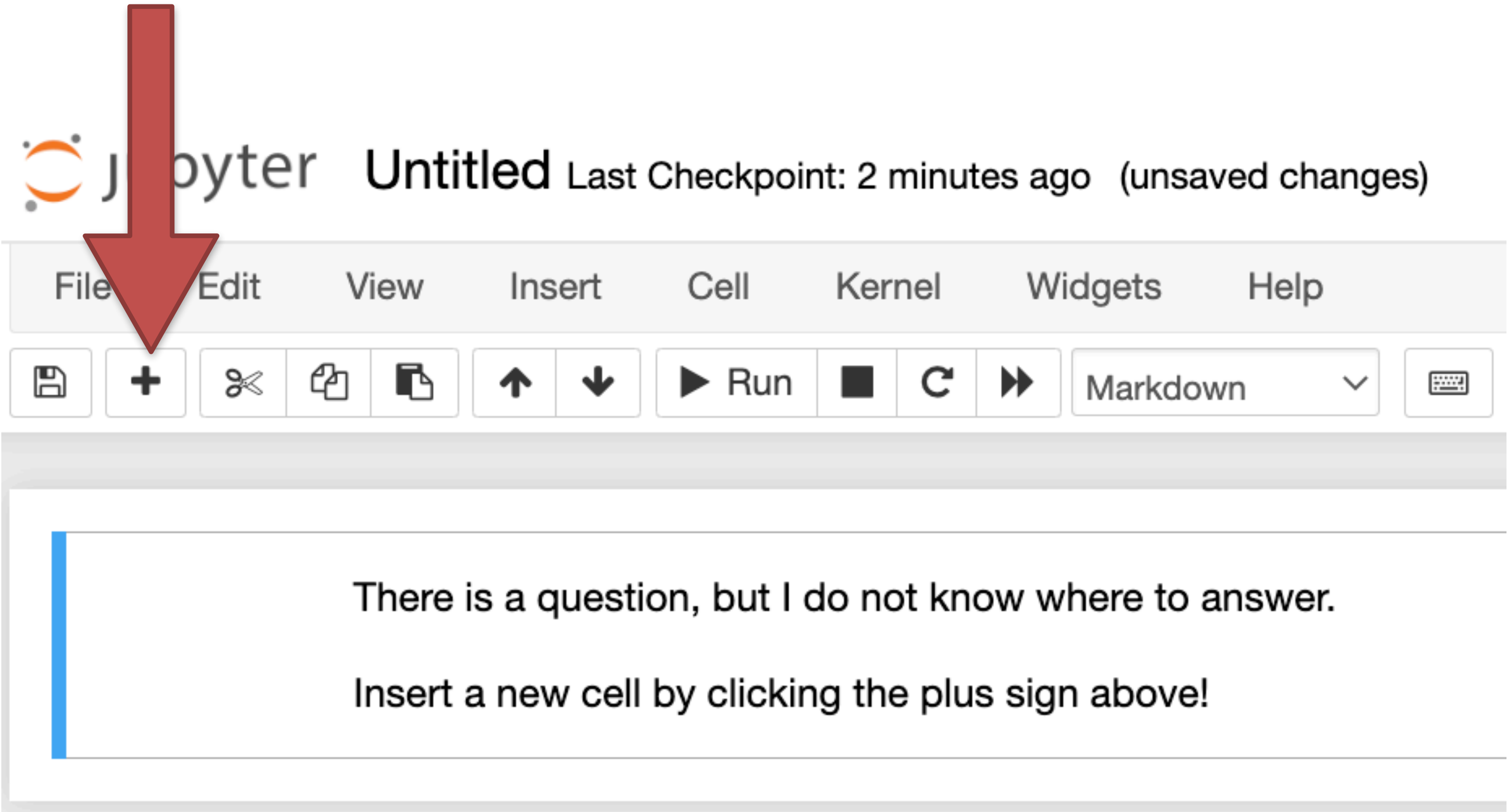
(make sure you know how to create a new cell!)



EXAM INFO

- Each section is worth 22-33 points (this may change)
- All 5 sections will be graded
- The questions that you must answer (or attempt to answer) will be indicated on the next slide
- For the rest, choose what you know best
- ~~Maximum 100 points (even if you get max of 145)~~
- ~~Grade 'A' is for 92-100 points~~
- 1.5 hour only
- If anything is unclear, you can ask questions during the exam

EXAM INFO

The questions that you must answer (or attempt to answer)

- Dynamic programming calculation
- SP calculation
- BLAST extension calculation
- BLAST alignment analysis
- One pseudocode question (!)
- NextGene questions
- FC (fold change) and FDR filtering

Expected to know

- What are NexGen sequencers?
- Names of at least two platforms (e.g. Illumina, Ion Torrent, Nanopore, PacBio)

Not expected to know

- Differences between instruments, except: Illumina and Ion Torrent are short reads, Nanopore and PacBio are long reads
- Names of instruments
- How sequencing works

Expected to know

- Why is sequence similarity important? (Shared information content: structure, function, history)
- Different types of evolutionary relationships (e.g. homologs, paralogs, orthologs)
- Why are dot plots useful? (Qualitative only)
- If you have a choice b/w using protein or DNA sequences when comparing genes, what would you use (Protein - why?)
- Dot Plot / LCS Plot algorithms (nested for-loops with if-statements) - no programming necessary

Expected to know

- Why is sequence similarity important? (Shared information content: structure, function, history)
- Different types of evolutionary relationships (e.g. homologs, paralogs, orthologs)
- Why are dot plots useful? (Qualitative)
- If you have a choice b/w using protein or DNA sequences when comparing genes, what would you use (Protein - why? Higher sequence conservation compare to DNA sequence)
- Dot Plot / LCS Plot algorithms (nested for loops with if statements) - no programming necessary

Expected to know

The idea behind dynamic programming

The Needleman-Wunsch algorithm, including:

- DP matrix initialization
- Filling in DP matrix
- Trace back
- Generating alignment

Expected to know

- Smith-Waterman local alignment algorithm
- What are BLOSUM and PAM, when/why do we use them?

(Scoring matrices for protein sequence alignment)

- What are affine gaps? (higher penalty for gap opening, lower penalty for gap extension. Affine = stretching. Chemical affinity ≠ affine gaps). Why did we need splice-tolerant aligners if NGS algorithms can already handle gaps?

Not expected to know

How to build substitution matrices

Common errors:

BLOSUM and PAM are NOT for nucleotide sequences!

Expected to know

- What is the meaning of "probability of an event"?
- What is the meaning of "P-value for event"?
- How do you use a P-value? (anything below cutoff is significant)
- Why is an amino acid match less frequent than a nucleotide match? (more amino acids)
- Differences in meaning of basic combinatorics rules (multiplication, permutation, combination) in application to sequence data. E.g. how many codons are possible?

Not expected to know

No distributions, no advanced math (*e.g.* no Bayes)

Expected to know

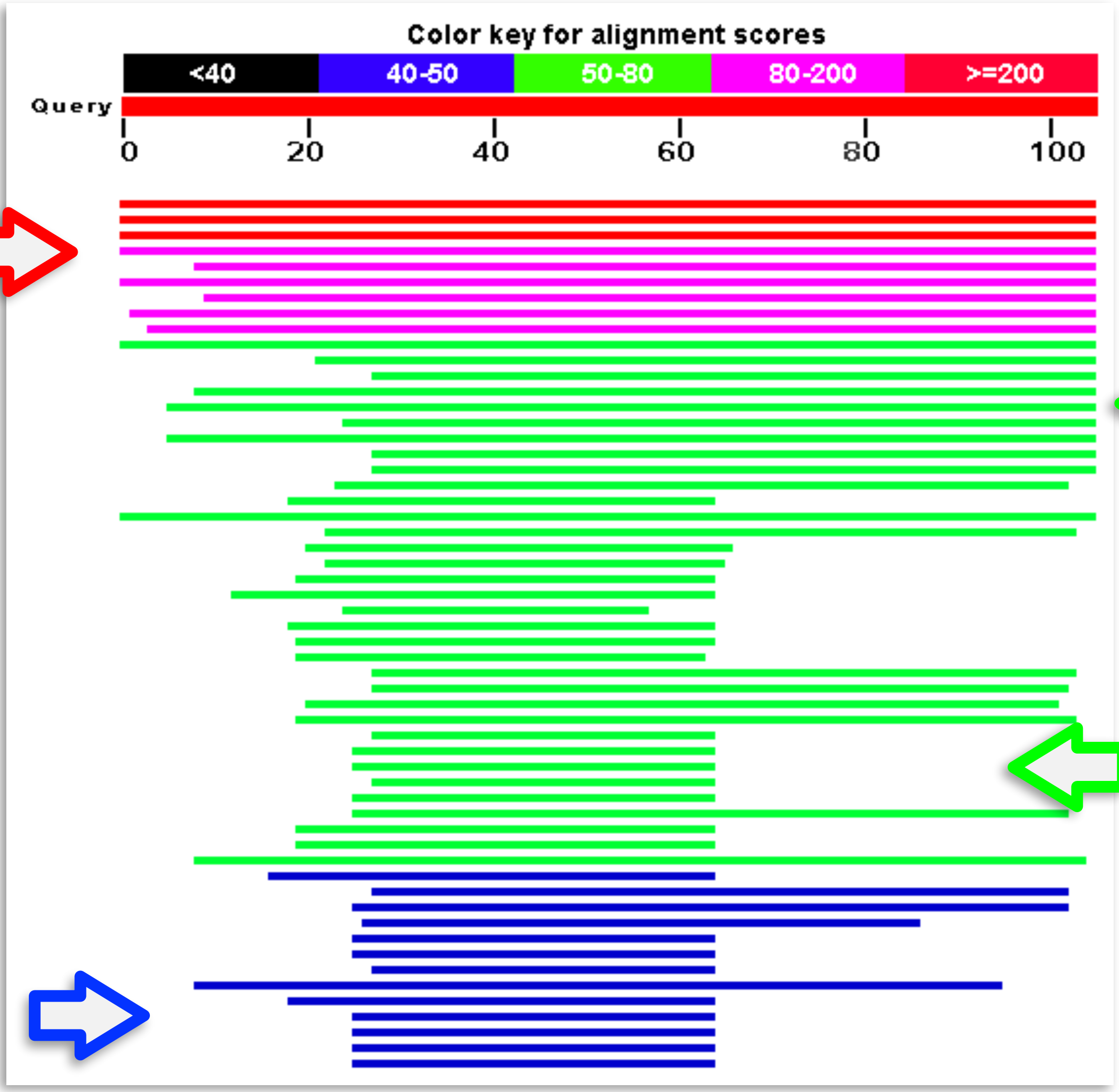
- Why BLAST?
- How does BLAST work? (basic idea)
- BLAST output statistics (e-value is the most important parameter!)
- BLAST output details (self-hit, paralogs, distant homologs, etc.)
- Relationship between E-value and P-value (e-value < 0.01 ~ p-value)
- e-value < 0.005 is trustable (rule of thumb)

Not expected to know

How to calculate E-value

BLAST output: visualized

high similarity (similarity is determined by score)



high coverage

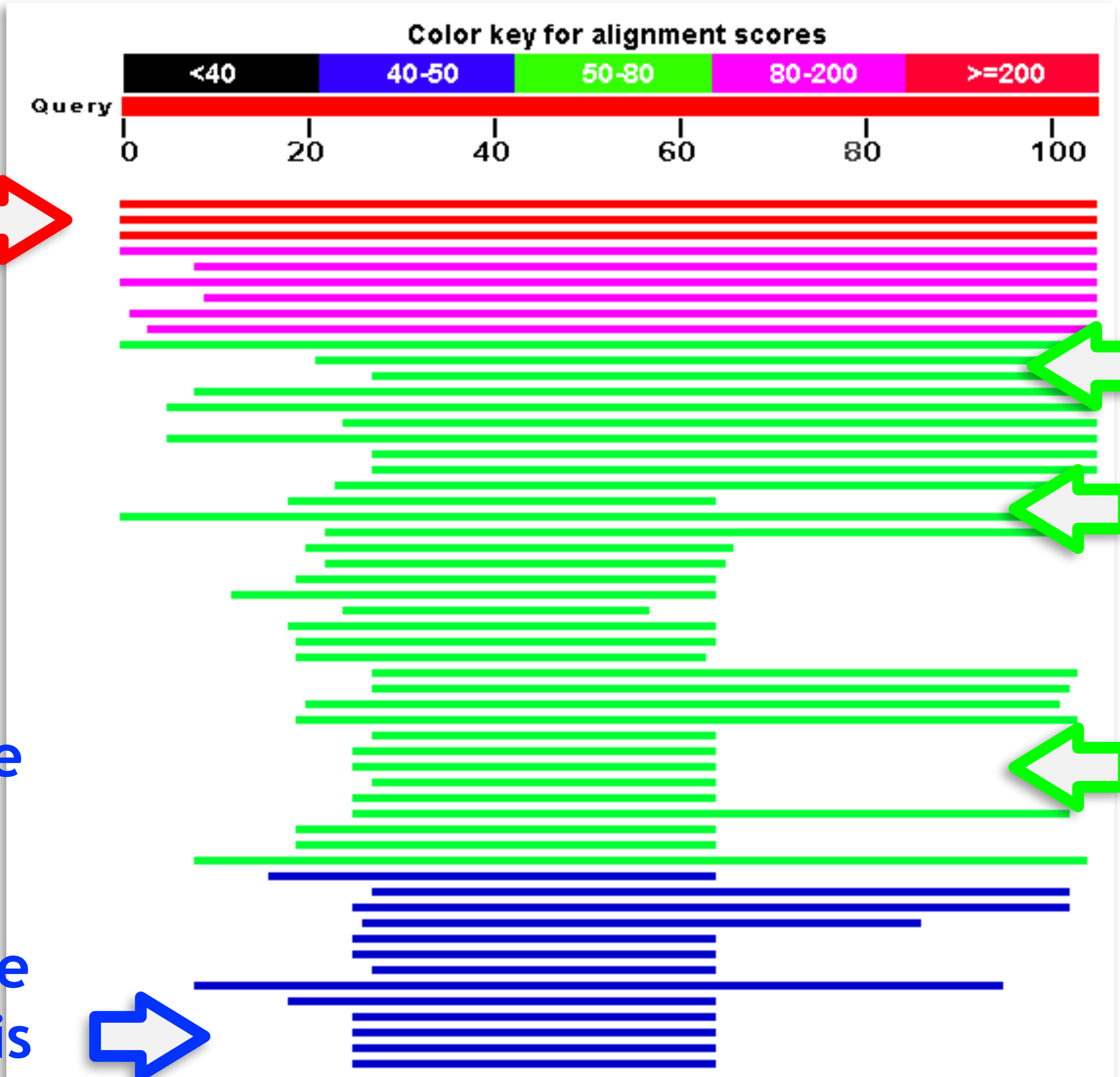
low coverage (coverage is determined by length)

low similarity

Overall score is sum of scores of individual aligned residues (aa or nt)

Coverage: visualized (MUST answer!) BLAST

Self-hit
Isoforms
Orthologs
Paralogs



Orthologs
Paralogs

Pseudogenes
(Free to mutate along entire length)

Functional domain

Dubious, but if you see multiple sequences with the same start/end, it is a domain

Expected to know

- Dictionary approach for finding seeds
- Importance of seed preprocessing
- Seed extension for HSPs
- What is a heuristic? Why is BLAST heuristic?

  (because:calculated guesses, shortcuts, not perfect)
- What do we consider when tuning seed length?

  (sensitivity and specificity)

Not expected to know

“modern” BLAST details (e.g. PSI-BLAST and PSSMs)

Expected to know

- What Bowtie is used for (align many reads against a single reference genome)
- When to use Bowtie vs. BLAST
- Advantages/Disadvantages of Bowtie vs BLAST

Not expected to know

- Burrows-Wheeler Transform
- Suffix trees and string searching

BLAST

Pros
- More Sensitive
- Online tool access
- Enormous sequence database

Cons
- Slower
- More false positive mappings

BWT-based alignment

Pros
- Fast
- Memory efficient

Cons
- Less sensitive
- No easy web interface
- Not in protein space (yet?)

There is no silver bullet:

The best tool is the right tool for the job

Expected to know

- Why is DP good for MSA? (Exhaustive algorithm)
- Why is DP a bad idea for MSA? (Will take impossibly long time)
- What is the basic idea behind progressive alignment? (Pairs first)
- How does Clustal work? (Progressive alignment with guide tree, Sum of Pairs scores)
- Calculating SP scores

Not expected to know

- Filling in a 3D DP matrix
- Filling in a 2D DP SP (Sum of Pairs) matrix
- DP computational complexity metrics
- Details of Feng-Doolittle algorithm

Expected to know

- How are MSA and phylogenetic trees related? (MSA is used to build trees in max parsimony)
- Distance-based versus enumerative methods: pros and cons

(Distance faster, but enumerative less error-prone)

- UPGMA algorithm (will not be required to build a tree)
- NJ concepts; why is NJ better than UPGMA? (no molecular clock)
- Picking the most parsimonious tree (know the concept but not enumeration)

Not expected to know

- Full UPGMA algorithm
- Full NJ algorithm
- How to enumerate all possible trees in max parsimony

Next Generation Sequencing

- NextGen and RNA-Seq terms
  - The purpose is to review the terms while answering the exam questions
  - Open book
  - About 10 terms out of the following slide
  - Several abbreviations that you have not encountered (look for it on Internet or on chatGPT)
- Coverage
  - The question will be exactly as in the slide, but with different numbers
- Single-end vs. paired-end reads
  - Just explain the difference and explain why paired-end reads are better

Very incomplete glossary of NextGen terms Lecture 18

- Easiest points on the open-book exam! Do not ignore!
- Some terms you might have to look up on the web

Term	Short Definition
AWS	Amazon Web Services
BAM	Binary Alignment/Map (complement to SAM)
BED	Browser Extensible Data (genomic coordinates)
bedtools	"a powerful toolset for genome arithmetic"
CIGAR	Concise Idiosyncratic Gapped Alignment Report
FASTA	FAST-All (biological sequence with a name)
FASTQ	Sequence with a Quality header and quality score
GATK	Genome Analysis Tool Kit
GFF	General Feature Format (gene metadata)
GTF	Gene transfer format (gene metadata)
IGV	Integrative Genomics Viewer (visual tool from Broad)
PHRED	Phil's Read Editor
rMATS	robust Multivariate Analysis of Transcript Splicing
RSEM	RNA-Seq by Expectation-Maximization
SAM	Sequence Alignment/Map
VCF	Variant Call Format
WES	Whole exome sequencing
WGS	Whole genome sequencing

Very incomplete glossary of NextGen terms Lecture 18

- Easiest points on the open-book exam! Do not ignore!
- Some terms you might have to look up on the web

Total 16 points.

Caveats this year:

Minus 2 points for each of the following two conditions even if all definitions are correct.

1. The Jupyter cell should be in ‘code’ format
2. Each term should be a variable and definition should be a string

E.g.
FASTQ = "Text-based format for storing both nucleotide sequences and quality scores from sequencing.”

Not
FASTQ - Text-based format for storing both nucleotide sequences and quality scores from sequencing.

Rule of Thumb: 30x coverage at a single location is enough to confidently call a SNP at that location.

How many 100 bp reads would you need to get an average of 30x coverage across the genome? (You should know the size of human genome, ca. 3 GB).

$$\text{Number of reads} = \text{Coverage} \times \frac{\text{Length of Genome}}{\text{Read Length}}$$

$$\text{Number of reads} = 30 \times \frac{3{,}000{,}000{,}000}{100\text{ bp}}$$

$$\text{Number of reads} = 900{,}000{,}000$$

Example: dividing length of genome by read length

Length of Genome / Read Length

What will you get?

$$\text{Number of reads} = \text{Coverage} \times \frac{\text{Length of Genome}}{\text{Read Length}}$$

Length of Genome / Read Length will get you what?
– Number of reads. **With what coverage?**

$$\text{Number of reads} = \text{Coverage} \times \frac{\text{Length of Genome}}{\text{Read Length}}$$

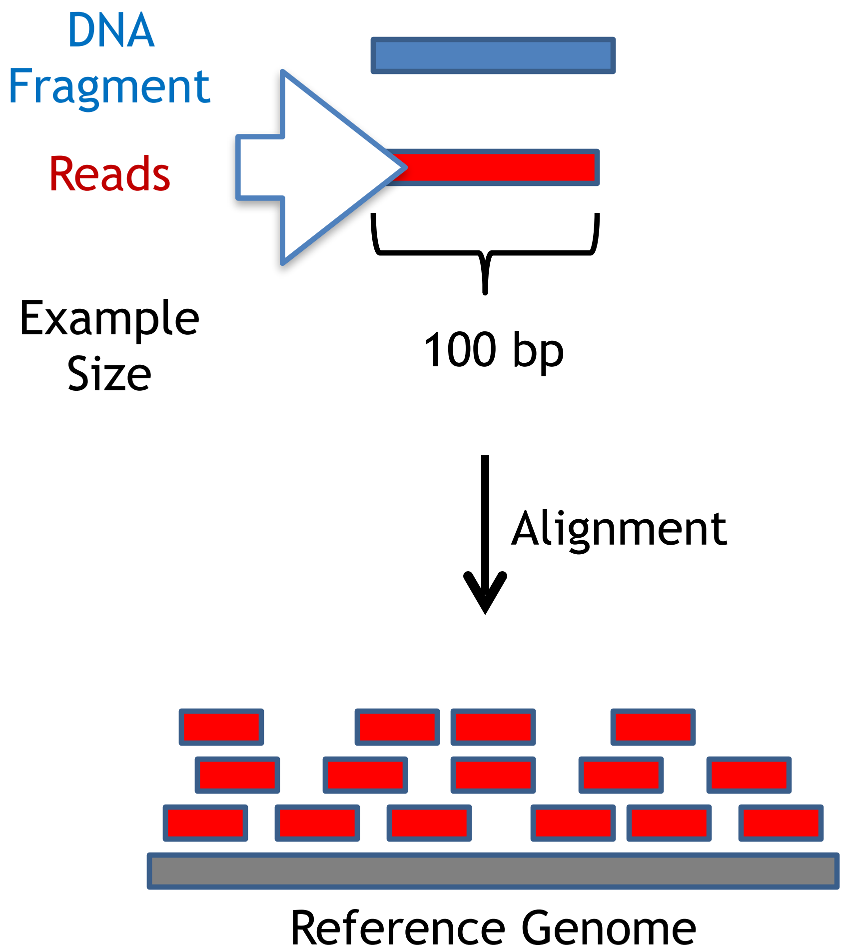
Length of Genome / Read Length will get you what?
– Number of reads. **With what coverage?**
Coverage of 1.

$$\text{Number of reads} = \text{Coverage} \times \frac{\text{Length of Genome}}{\text{Read Length}}$$

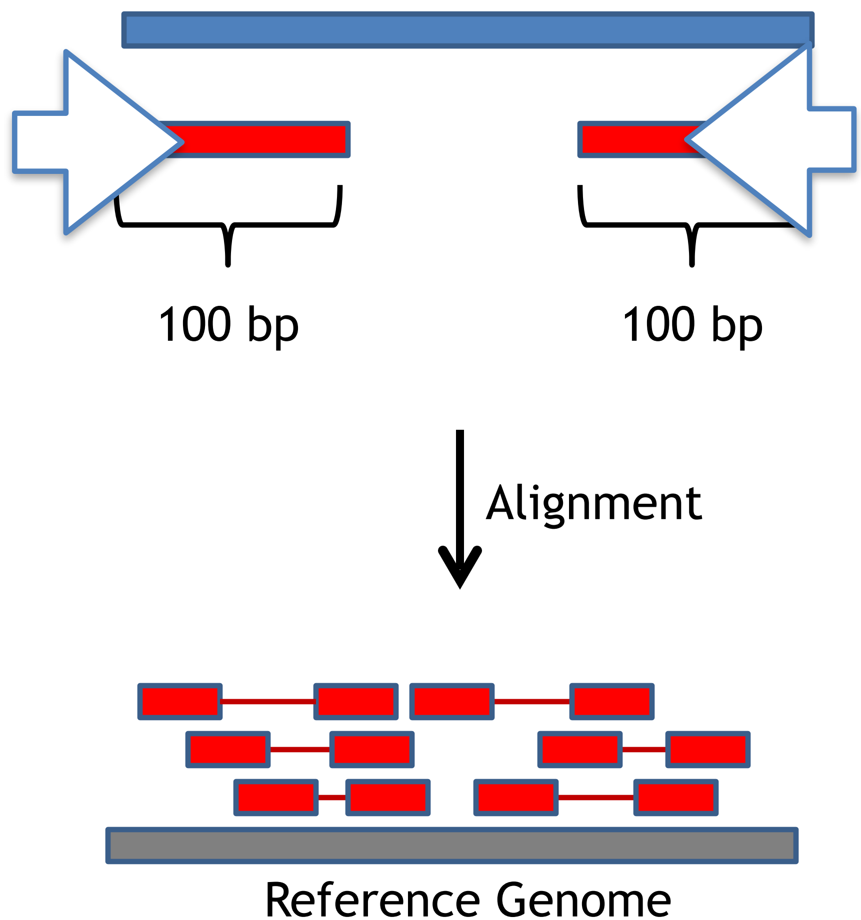
$$\text{Number of reads} = 1 \times \frac{3{,}000{,}000{,}000}{100\ \text{bp}}$$

$$\text{Number of reads} = 30{,}000{,}000$$

Single end reads



Paired end reads



Single End vs. Paired End

'single end': one end of the DNA molecule (of a read).
'paired ends': two ends of the same DNA molecule (of a read).

"Sequence one end, turn it around and sequence the other end."

Why paired ends are better than single end reads:
You can use the information about the distance between the ends to map to a reference genome (or assemble the sequence de-novo)

Paired end reads resolve structural rearrangements (insertions, deletions, inversions), and help mapping / assembling across repetitive regions.

Adopted from http://seqanswers.com/forums/showthread.php?t=503

Quality Control: Phred scores

Phred quality scores are logarithmically linked to error probabilities

quality = $-10*\log_{10}(P)$

Example: p-value = 0.001 (or 1/1000 or 10^{-3})

$Q = -10*\log_{10}(0.001) = -10*-3 = 30$

If Phred assigns a quality score of 30 to a base, the chances that this base is called incorrectly are 1 in 1000.

Phred Score	Probability of incorrect	Base call accuracy
10	1 in 10	90%
20	1 in 100	99%
30	1 in 1000	99.9%
40	1 in 10,000	99.99%
50	1 in 100,000	99.999%
60	1 in 1,000,000	99.9999%

Quality Control: Phred scores

Phred quality scores are logarithmically linked to error probabilities

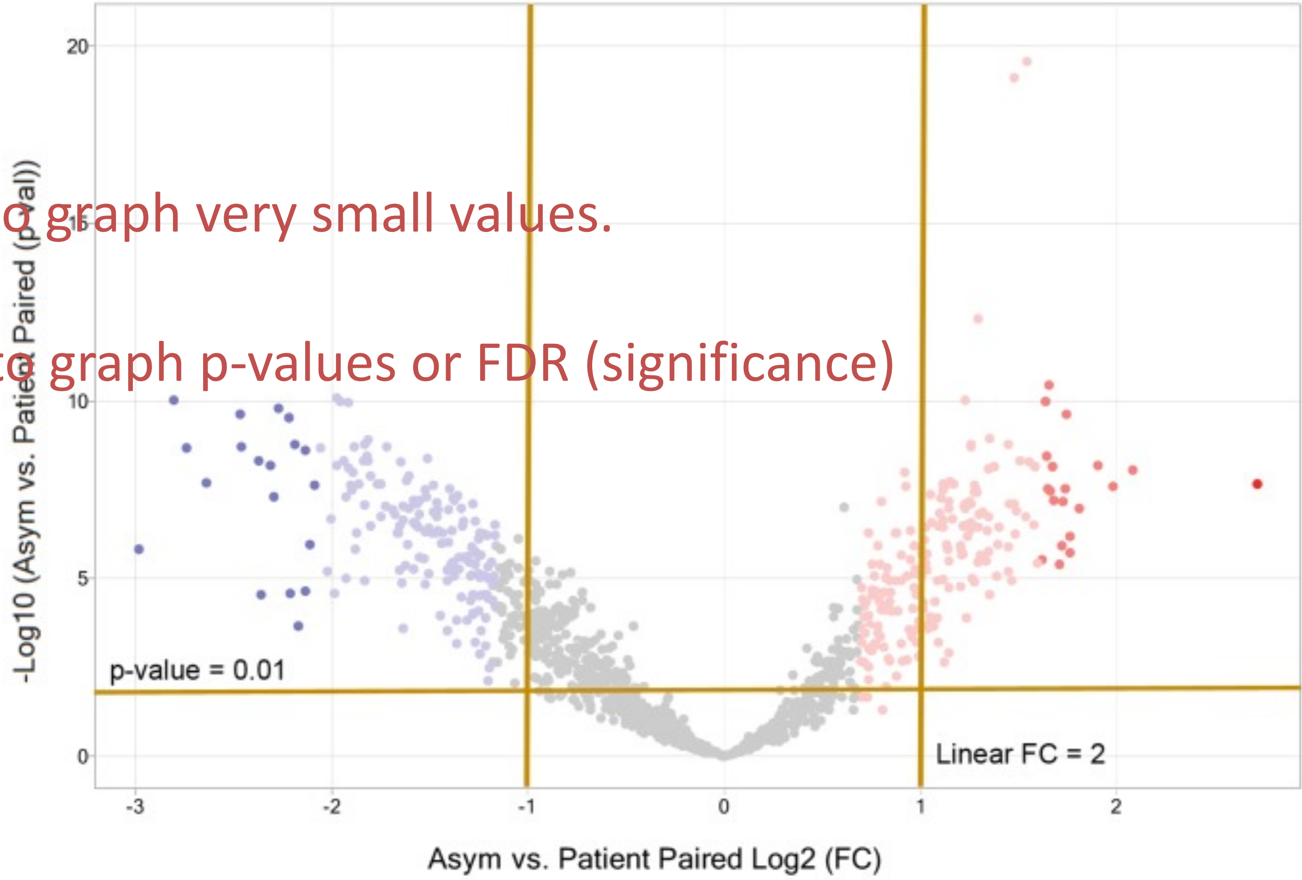
PHRED stands for Phil's Read Editor

quality = $-10*\log_{10}(P)$

Why $-10*\log_{10}(P)$?

Same on Volcano plot: to graph very small values.

$-10*\log_{10}$ is often used to graph p-values or FDR (significance)



Quality scores

ASCII	Hex	Symbol	ASCII	Hex	Symbol	ASCII	Hex	Symbol	ASCII	Hex	Symbol
32	20	(space)	48	30	0	64	40	@	80	50	P
33	21	!	49	31	1	65	41	A	81	51	Q
34	22	"	50	32	2	66	42	B	82	52	R
35	23	#	51	33	3	67	43	C	83	53	S
36	24	$	52	34	4	68	44	D	84	54	T
37	25	%	53	35	5	69	45	E	85	55	U
38	26	&	54	36	6	70	46	F	86	56	V
39	27	'	55	37	7	71	47	G	87	57	W
40	28	(	56	38	8	72	48	H	88	58	X
41	29	)	57	39	9	73	49	I	89	59	Y
42	2A	*	58	3A	:	74	4A	J	90	5A	Z
43	2B	+	59	3B	;	75	4B	K	91	5B	[
44	2C	,	60	3C	<	76	4C	L	92	5C	\
45	2D	-	61	3D	=	77	4D	M	93	5D	]
46	2E	.	62	3E	>	78	4E	N	94	5E	^
47	2F	/	63	3F	?	79	4F	O	95	5F	_

What is the probability of error?

```
@HWUSI-EAS1671_0006:1:1:1031:5138#0/1
TNAGATGAAGCACTGTAGCTATCCCGTATGCCGTC
+HWUSI-EAS1671_0006:1:1:1031:5138#0/1
[BZT_]YT]^cYc\\c``Yc\\^M\QTVVTccTcc
```

Header

Sequence

Quality Header

Quality Scores

Phred value + 33 = decoding of the ascii character

Quality scores

ASCII	Hex	Symbol	ASCII	Hex	Symbol	ASCII	Hex	Symbol	ASCII	Hex	Symbol
32	20	(space)	48	30	0	64	40	@	80	50	P
33	21	!	49	31	1	65	41	A	81	51	Q
34	22	"	50	32	2	66	42	B	82	52	R
35	23	#	51	33	3	67	43	C	83	53	S
36	24	$	52	34	4	68	44	D	84	54	T
37	25	%	53	35	5	69	45	E	85	55	U
38	26	&	54	36	6	70	46	F	86	56	V
39	27	'	55	37	7	71	47	G	87	57	W
40	28	(	56	38	8	72	48	H	88	58	X
41	29	)	57	39	9	73	49	I	89	59	Y
42	2A	*	58	3A	:	74	4A	J	90	5A	Z
43	2B	+	59	3B	;	75	4B	K	91	5B	[
44	2C	,	60	3C	<	76	4C	L	92	5C	\
45	2D	-	61	3D	=	77	4D	M	93	5D	]
46	2E	.	62	3E	>	78	4E	N	94	5E	^
47	2F	/	63	3F	?	79	4F	O	95	5F	_

What is the probability of error?

Phred value + 33 = decoding of the ascii character

Q = 63-33 = 30

from a previous slide:

$Q = -10*\log_{10}(0.001) = -10*-3 = 30$

Probability of error = 0.001

Alignment Results

Reference Genome

```
coor     12345678901234  5678901234567890123456789012345
ref      AGCATGTTAGATAA**GATAGCTGTGCTAGTAGGCAGTCAGCGCCAT
```

Read Alignments

```
r001+          TTAGATAAAGGATA*CTG
r002+         aaaAGATAA*GGATA
r003+       gcctaAGCTAA
r004+                     ATAGCT..............TCAGC
r003-                           ttagctTAGGC
r001-                                       CAGCGCCAT
```

Sequence Alignment/Map (SAM) file format

Header

Read Alignments

```
@SQ SN:ref LN:45
r001 163 ref  7 30 8M2I4M1D3M = 37  39 TTAGATAAAGGATACTA *
r002   0 ref  9 30 3S6M1P1I4M * 0    0 AAAAGATAAGGATA    *
r003   0 ref  9 30 5H6M       * 0    0 AGCTAA    *   NM:i:1
r004   0 ref 16 30 6M14N5M    * 0    0 ATAGCTTCAGC       *
r003  16 ref 29 30 6H5M       * 0    0 TAGGC     *   NM:i:0
r001  83 ref 37 30 9M         = 7  -39 CAGCGCCAT         *
```

Read ID

Chromosome

Position

CIGAR string

Aligned Sequence

Mismatches

- **CIGAR** stands for

Concise Idiosyncratic Gapped Alignment Report.

- Idiosyncratic: “specific” or “unique” in this context
- Compressed representation of an alignment used in the SAM format.

For example:

RefPos: 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19

Reference: C C A T A C T G A A C T G A C T A A C

Read: ACTAGAATGGCT

Aligning these two:

http://bioinformatics.cvr.ac.uk/blog/tag/cigar-string/

CIGAR: Alignment QC Compression

CIGAR stands for
Concise Idiosyncratic Gapped Alignment Report.
It is a compressed representation of an alignment that is used in the SAM file format.

RefPos:	1	2	3	4	5	6	7		8	9	10	11	12	13	14	15	16	17	18	19
Reference:	C	C	A	T	A	C	T		G	A	A	C	T	G	A	C	T	A	A	C
Read:	-	-	-	-	A	C	T	A	G	A	A	-	T	G	G	C	T	-	-	-
Match index					1	2	3		1	2	3		1	2	3	4	5			
Match tally					M	M	M		M	M	M		M	M	M	M	M			
Insertion index								1												
Insertion tally								i												

CIGAR: 3M 1I 3M 1D 5M

3(three)M(atches, then)1(one)I(sertion, then)3M(atches)1D(eletion, then)5M(atches)

3M1I3M1D5M

Both insertion and deletion refer to the reference genome:
insertion compared to the genome, deletion compared to the reference genome

OMICS and RNA-Seq

- OMICS and RNA-Seq terms
  - Open book
  - The terms that you might encounter with vendor data

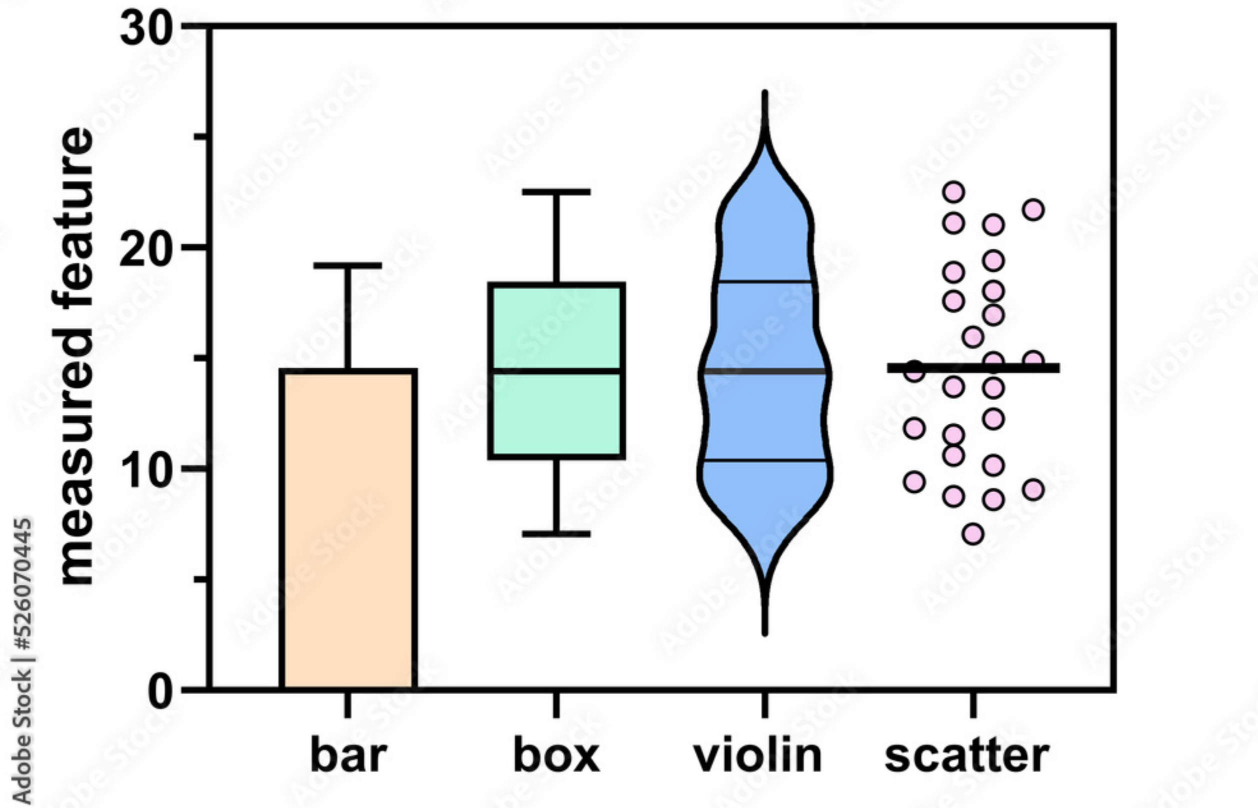
Term	Short Definition
FC	Fold Change
FDR	p-value, adjusted for multiple testing. False Discovery Rate,
p_adj	p-value, adjusted for multiple testing
Benjamini	p-value, adjusted for multiple testing. Benjamini and Hochberg correction.
BH	p-value, adjusted for multiple testing. Benjamini and Hochberg correction.

- Understanding statistical tests
  - T-test
  - Types of plots
  - Why both Fold Change (FC) and significance (p-value or FDR) are necessary to call the gene 'differentially expressed'
  - On board: t-test examples

A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S
	Normal mRNA copies									Cancer mRNA copies							alpha =	0.05
expression of	Samples (replicates): S1, S2								expression of	replicates: S1, S2								
gene:	S1	S2	S3	S4	S5	Mean	Stdev	T-test	gene:	S6	S7	S8	S9	S10	Mean	Stdev		
gene1	10	15	25	13	15	15.6	5.6	0.002	gene1	20	30	40	25	20	27	8.4	significant?	yes
gene2	10	1	39	13	15	15.6	14.1	0.051	gene1	20	30	50	25	10	27	14.8	significant?	no
g3									g3									

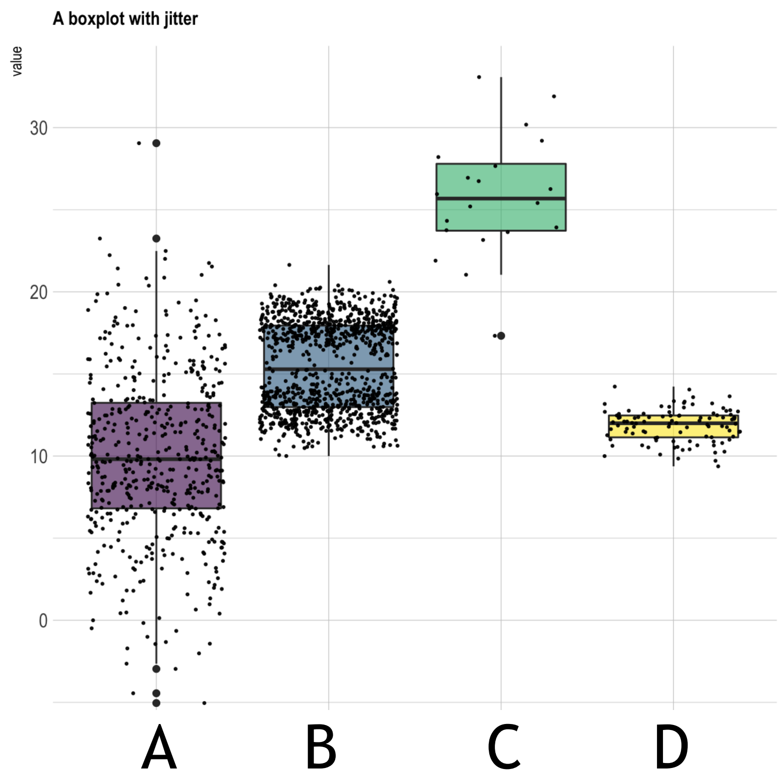
https://docs.google.com/spreadsheets/d/1A8sfa0mdVC_9k-ruDeZayfsqrvZd5CWQWyu9xKx0OmY/edit?usp=sharing

- Avoid bar plots even with standard deviations: bar plots do not show the distribution of replicates. Outliers can distort the results without being apparent
- Violin or box-scatter plots are best



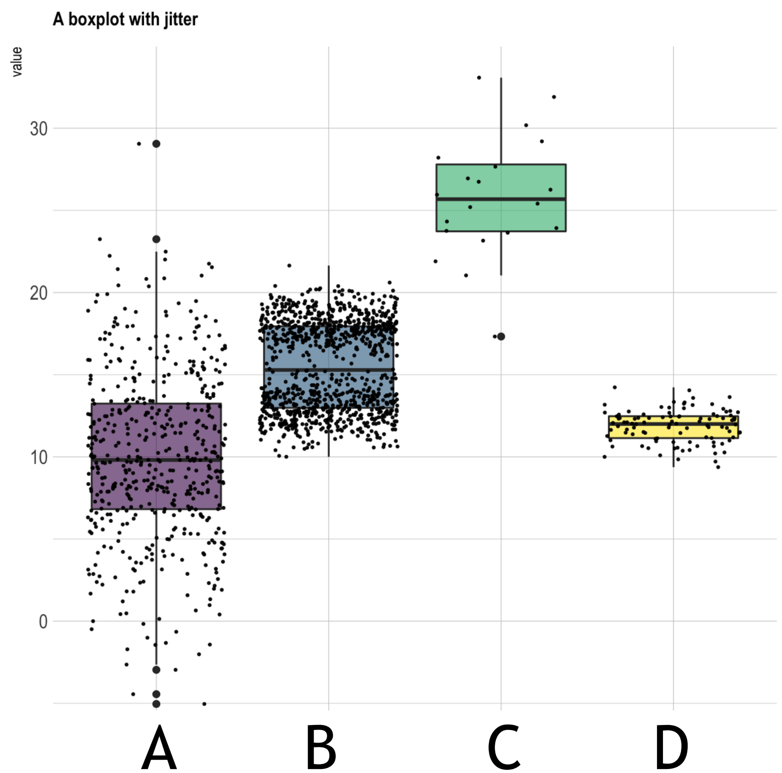
https://stock.adobe.com/images/an-illustration-comparing-different-types-of-data-representation-a-standard-box-plot-orange-bars-and-whiskers-green-violin-plot-blue-and-scatter-plot-purple/526070445

- Use the plots to QUALITATIVELY evaluate whether gene is differentially expressed. Is this gene differentially expressed in conditions A, B, C, and D? Each point (each dot) is a replicate.



https://r-graph-gallery.com/89-box-and-scatter-plot-with-ggplot2.html

- Use the plots to qualitatively evaluate whether gene is differentially expressed.



- Are these genes differentially expressed?
- A vs. B vs. C - yes, C vs. D - yes, B vs. D - probably, A vs D - no.

- Limited to understanding of data filtering based on FC, FDR, and absolute expression value (TPM = Transcripts Per Million)
- Example:

If the thresholds for differential expression are

|FC| > 1.2 and FDR < 0.1 (what do pipes around FC mean?)

which of the genes would you consider to be differentially expressed based on the following table?

The genes should pass both FC and FDR thresholds.

Gene	FC	FDR
Gene1	1.6	0.05
Gene2	-2	0.01
Gene3	1.5	0.35
Gene4	1.1	0.2

Answer:

FC is Fold Change

FDR = False Discovery Rate = p-value adjusted for multiple correction

|FC| > 1.2 (absolute value) means FC > 1.2 or FC < -1.2. FC should be either greater than 1.2 or less than -1.2.

The genes that pass both FC and FDR < 0.1 thresholds are

- gene1 with FC = 1.6, FDR = 0.05
- gene2 with FC = -2, FDR = 0.01

Gene 3 passes FC threshold, but does not pass FDR, and Gene4 does not pass either threshold.

Gene	FC	FDR
Gene1	1.6	0.05
Gene2	-2	0.01
Gene3	1.5	0.35
Gene4	1.1	0.2

- Example:

If the thresholds for differential expression are

|FC| > 1.5, FDR < 0.1, TPM > 5

which of the genes would you consider to be differentially expressed based on the following table?

The genes should pass FC, FDR, and TPM thresholds.

Gene	log2FC	FDR	average TPM
Gene1	0.68	0.05	2
Gene2	-1.00	0.01	55
Gene3	0.58	0.35	1000
Gene4	0.14	0.2	1000

Answer:

Gene	log2FC	FDR	average TPM
Gene1	0.68	0.05	2
Gene2	-1.00	0.01	55
Gene3	0.58	0.35	1000
Gene4	0.14	0.2	1000

Note that FC is log2 here. Need to convert to unlogged FC.

Gene1: FC = 2^log2FC for positive logarithms. 2^0.68 = 1.6.

Gene1 passes FC threshold (FC>1.5) and FDR threshold (FDR < 0.1), but not TPM threshold (TPM > 5).

Answer:

Gene	log2FC	FDR	average TPM
Gene1	0.68	0.05	2
Gene2	-1.00	0.01	55
Gene3	0.58	0.35	1000
Gene4	0.14	0.2	1000

Gene	FC	FDR
Gene1	1.6	0.05
Gene2	-2	0.01
Gene3	1.5	0.35
Gene4	1.1	0.2

Note that FC is log2. Need to convert to unlogged FC.

Gene1: FC = 2^log2FC for positive logarithms. 2^0.68 = 1.6.

Gene1 passes FC threshold (FC>1.5) and FDR threshold (FDR < 0.05), but not TPM threshold (TPM > 5).

Different procedure for negative logs:

Gene2: FC = -1/(2^log2FC) for negative logarithms. -1/(2^-1) = -1/0.5 = -2

Gene2 passes FC threshold (FC<-1.5), FDR threshold (FDR < 0.1), and TPM threshold (TPM > 5).

Gene3 passes FC and TPM thresholds, but does not pass FDR

Gene4 passes TPM threshold, but does not pass FC and FDR

Expected to know
Why we used a particular tool in lab

Not expected to know
The precise details of its arguments, formats, etc.

<u>Example Question</u>
What would be the fastest way to look for local alignments between all genes in mouse and all genes in human?

<u>Valid Answer</u>
For ultimate speed we could do a completely offline BLAST of mouse sequences against a human database on our computer.

Lab 4 (must answer)
How to fill in a DP matrix with global or local algorithms

Lab 7 (must answer)
Online BLAST: how to read and interpret BLAST results

Lab 11
Different online MSA tools, e.g. compare Clustal to T-COFFEE, etc. (ClustalW is faster, T-Coffee is more accurate)

Pseudocode

"There are only 10 types of people in the world: those who understand binary and those who don't."

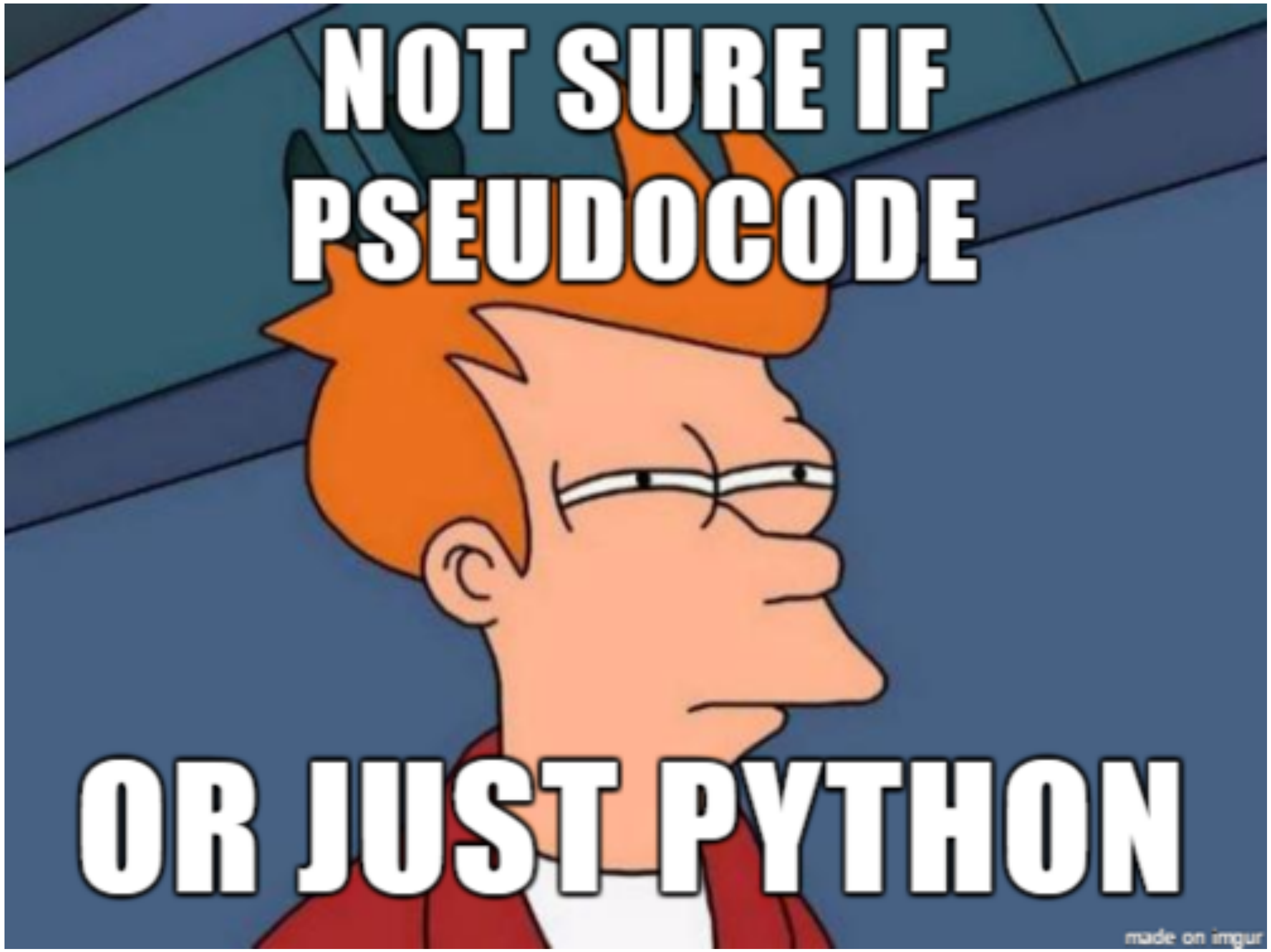
Pseudocode

(Just for the reference)

Only one question.
Do not have to follow formal pseudocode rules.

What Is Pseudocode?

- An implementation of an algorithm in the form of annotations and informative text written in plain English.
- Pseudocode has no syntax like any of the programming language
- Pseudocode can’t be compiled or interpreted by the computer.


NOT SURE IF
PSEUDOCODE
OR JUST PYTHON
made on imgur

Why Pseudocode?

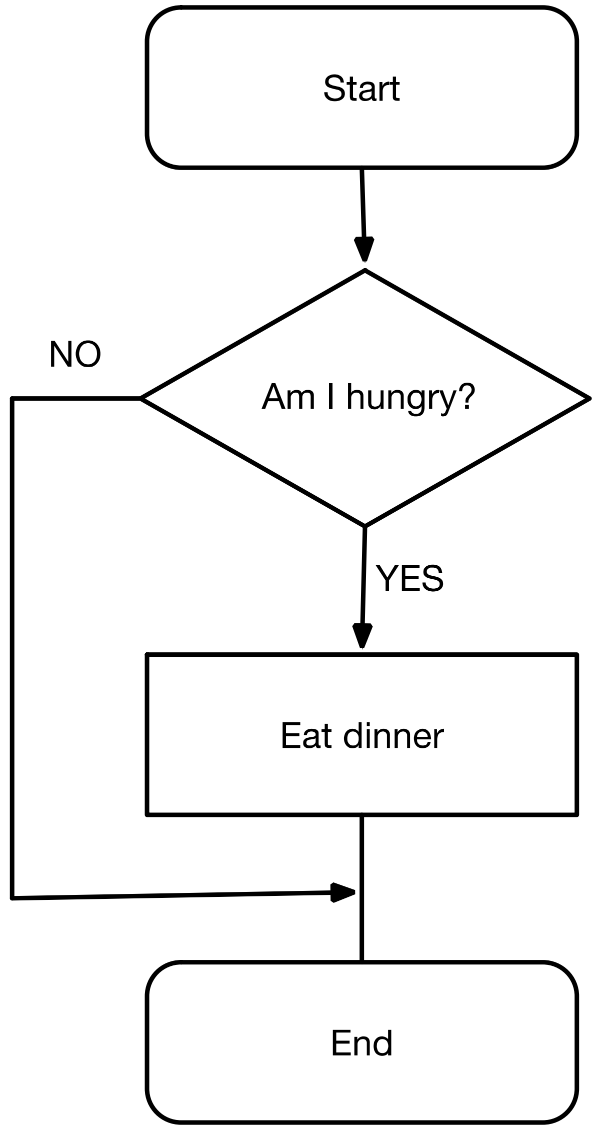
- Programming languages are difficult to read for most people, but **pseudocode** allows non-programmers, such as business analysts, to review the steps to confirm the proposed code matches the coding specifications.

- By writing the code in human language first, the programmer safeguards against leaving out an **important** step.

How To Make A Peanut Butter Sandwich

- Open drawer
- Find a knife in the drawer
- Take knife out of drawer and put on counter
- Take bread out
- Take 2 slices from the pack and place on counter
- Open the fridge
- Take peanut butter out of the fridge and put on counter
- Spread the peanut butter on one side with knife
- Put slice of bread without peanut butter on top of the slice with peanut butter
- Put peanut butter back in fridge
- Eat

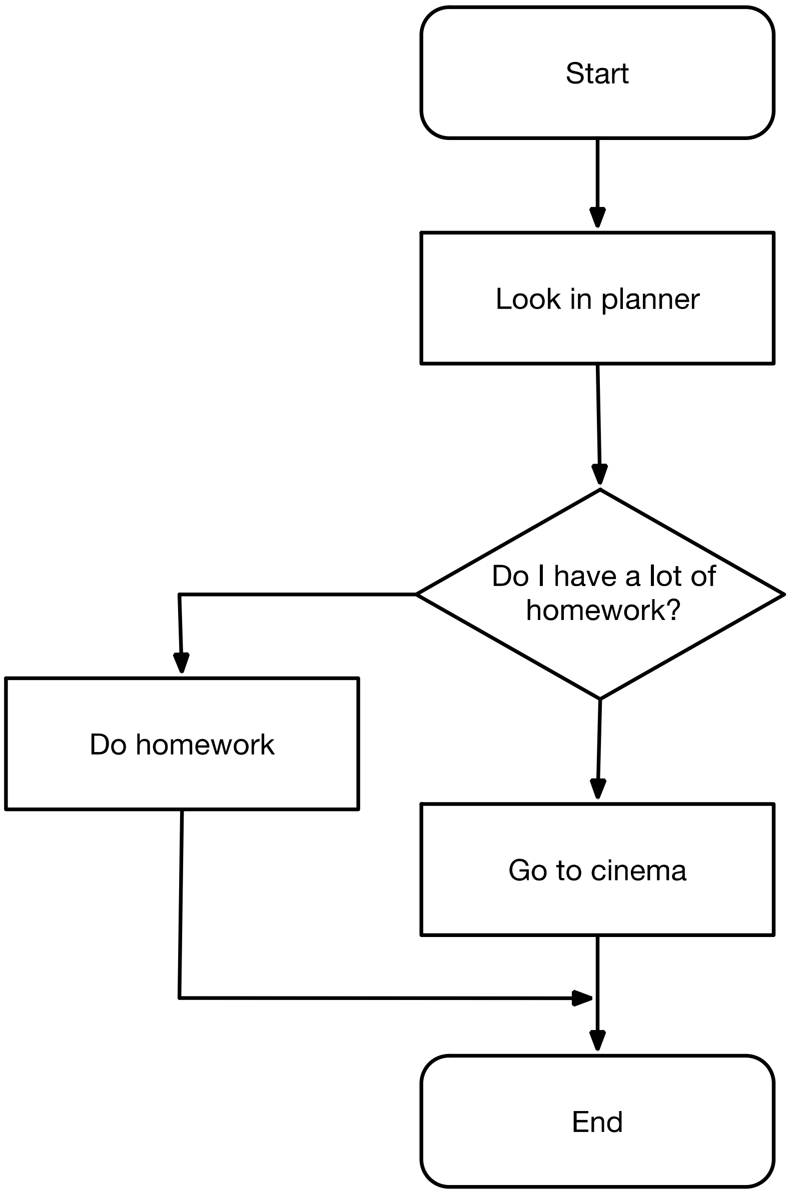
Flowchart



Pseudocode

IF I am hungry
THEN
    Eat dinner
ELSE
    Don't eat
END IF

Flow Chart



Pseudocode

Look in planner
IF I have lots of homework to do **THEN**
    Do homework
ELSE
    Go to cinema
END IF

How To Make A Peanut Butter Sandwich

IF drawer is closed

    THEN open drawer

IF knife in drawer

    THEN OUTPUT knife

ELSE

    Look for knife

OUTPUT bread

OUTPUT 2 slices of bread and place on counter

IF fridge door closed

    THEN Open fridge

ELSE

    Leave it closed

IF you want peanut butter

    THEN OUTPUT peanut butter from fridge and put on counter

ELSE

    Leave it

OPEN peanut butter

    INPUT knife

    OUPUT peanut butter on knife

Spread peanut butter on 1 side of bread

IF slice of bread with no peanut butter

    THEN pick up and place on other slice of bread

ELSE

    Check other slice for peanut butter

IF you want sandwich cut in half

    THEN Pick up knife and cut in half

ELSE

    Leave whole

OPEN fridge door

    INPUT peanut butter to fridge

    CLOSE fridge door

IF ready to eat

    THEN Go and enjoy

ELSE

    Wait

Pseudocode: What is NOT Allowed?

AMBIGUOUS statements!

Pseudocode: What is Allowed?

Programming Statements

IF

- `if a is greater than b`
- `if character is U`

FOR

- `for each item in list`
- `for each character in string`
- `for each protein in genome`

WHILE

Pseudocode Example

```
IF student's grade is greater than or equal to 60
   print "passed"
ELSE
   print "failed"
```

Comment Pseudocode

Comment pseudocode using # or //!

and // are commenting characters.

Example 1: Reading FASTA File

```
algorithm: reading fasta file for each header and processing
it
input: fasta_file
-----------------------------------------------------------
initialize sequence_name as a string of characters
initialize sequence as an empty string of characters
for each line in fasta_file
       if line starts with '>'
              process sequence for sequence name
              save the line content following '>'
sequence_name
              reset sequence to empty
       else
              concatenate sequence with line
```

Example 1: Reading FASTA File

```
algorithm: reading fasta file for each header and
processing it
input: fasta_file
-------------------------------------------------------------
sequence_name <- ''
sequence <- ''
for each line in fasta_file
      if ( line[0] = '>' )
            process(sequence, sequence_name)
            sequence_name <- line[2:length(line)]
            sequence <- ''
      else
            sequence <- sequence + line
```

Equal Versus Assignment

Equal

Use **=** for comparison

compares a to value a

a = 1

Assignment

Use **<-** for assignment

compares **a** to value 1

a <- 1

Example 2: Orthologs

```
algorithm: finding all orthologs of proteins of genome1 in
genome2

input: genome1 : a file containing all proteins in fasta
format

              genome2 : name of an organism

output : list of ortholog proteins printed on screen
-----------------------------------------------------------

for each protein1 in genome1
       protein2 <- top result of blast(protein1, genome2)
       if (top result of blast(protein2, genome1) = protein1)
              print protein1 and protein2 are orthologs
```

Example 3: Finding the Maximum

```
algorithm: finding maximum of all numbers
input: list containing numbers
output: maximum value in list, in case of tie returns first

-----------------------------------------------------------
max <- -∞
for each number in list
       if number > max_result
              max <- number
return max
```

Example 3: Finding the Maximum

```
algorithm: finding index of maximum of all numbers
input: list containing numbers
output: index of maximum value in list, in case of tie returns first
-----------------------------------------------------------
max_index <- 1
N <- length of list
for i in range of 2 to N
        if list[i] > list[max_index]
                max_index <- i
return max_index
```

Best Pseudocode Practices

- Limit **pseudocode** statements to one per line.
- Use initial capitalization for all keywords.
- Do not write source code. Write your thoughts of what the program should do.
- List all steps. ...
- Do NOT be afraid to use programming structures (IF, THEN, ELSE etc.)
- Use indentation to show subordination.

On the board:
DP matrix

(must answer)

DP matrix

The dynamic programming table below was constructed from the DNA sequences GATTACA and GATAGA, with the scoring scheme match = 5, mismatch = 0, and gap = -3.

	–	C	A	C	G	A	C	A
–	0	0	0	0	0	0	0	0
C	0	2	0	0	0	0	0	0
A	0	0	4	1	0	2	0	2
T	0	0	1				2	0
A	0	0	2				2	4

This example is incorrect! It only shows the format that will be used at the exam.

Dynamic Programing

The idea is to solve a problem by solving its subproblems…

solution to problem of size **N** =
solution to problem of size **N-1**
+
some simple calculation

alignment of two sequences of length **N** =
alignment of two sequences **N-1**
+
picking match/mismatch or gap for the last character

Aligning X = GATTACA and Y = GATAGA — EXAMPLE

This is an example of a dynamic programming algorithm, which solves problems recursively by breaking them down.

$$S(i,j) = \max \begin{cases} S(i-1, j-1) & + c(X_i, Y_j) \\ S(i-1, j) & + gap \\ S(i, j-1) & + gap \end{cases}$$

Is it global or local alignment?

Aligning X = GATTACA and Y = GATAGA

EXAMPLE

This is an example of a **dynamic programming** algorithm, which solves problems recursively by breaking them down.

$$S(i,j) = \max \begin{cases} S(i-1, j-1) & + c(X_i, Y_j) \\ S(i-1, j) & + gap \\ S(i, j-1) & + gap \end{cases}$$

Is it global or local alignment?

-global, because there is no 0 (no forth term)

Scoring scheme

Match: +1

Mismatch: +0

Gap: -1

Aligning X = GATTACA and Y = GATAGA — EXAMPLE

	$X_i \rightarrow$	0	1	2	3	4	5	6	7
$Y_j \downarrow$		–	G	A	T	T	A	C	A
0	–								
1	G								
2	A								
3	T								
4	A								
5	G								
6	A								

Keep track of $S(i, j)$ in a table.

$S(i, j)$ is the score of the best alignment between $X_{1...i}$ & $Y_{1...j}$

Aligning X = GATTACA and Y = GATAGA

EXAMPLE

$X_i \rightarrow$		0	1	2	3	4	5	6	7
$Y_j \downarrow$		–	G	A	T	T	A	C	A
0	–	0							
1	G								
2	A								
3	T								
4	A								
5	G								
6	A								

DEFINITION:

$$S(0, 0) = 0$$

Aligning X = GATTACA and Y = GATAGA — EXAMPLE

$Y_j \downarrow$ / $X_i \rightarrow$		0	1	2	3	4	5	6	7
		–	G	A	T	T	A	C	A
0	–	0	→ -1	→ -2	→ -3	→ -4	→ -5	→ -6	→ -7
1	G								
2	A								
3	T								
4	A								
5	G								
6	A								

We fill in the top row by aligning Y_0 with a lot of **gaps**

E.g., $S(5,0) = -5 \rightarrow$

```
G A T T A
– – – – –
```

Aligning X = GATTACA and Y = GATAGA

EXAMPLE

$Y_j \downarrow$ / $X_i \rightarrow$		0	1	2	3	4	5	6	7
		–	G	A	T	T	A	C	A
0	–	0	-1	-2	-3	-4	-5	-6	-7
1	G	-1							
2	A	-2							
3	T	-3							
4	A	-4							
5	G	-5							
6	A	-6							

We fill in the left column by aligning $\mathbf{X_0}$ with a lot of **gaps**

How would you fill the top row and leftmost column in Smith-Waterman?

Aligning X = GATTACA and Y = GATAGA

EXAMPLE

$Y_j \downarrow$ / $X_i \rightarrow$		0	1	2	3	4	5	6	7
		–	G	A	T	T	A	C	A
0	–	0	-1	-2	-3	-4	-5	-6	-7
1	G	-1							
2	A	-2							
3	T	-3							
4	A	-4							
5	G	-5							
6	A	-6							

We fill in the left column by aligning X_0 with a lot of **gaps**

How would you fill the top row and leftmost column in Smith-Waterman? With all zeros.

Aligning X = GATTACA and Y = GATAGA **EXAMPLE**

	$X_i \rightarrow$	0	1	2	3	4	5	6	7
$Y_j \downarrow$		–	G	A	T	T	A	C	A
0	–	0	→ -1	→ -2	→ -3	→ -4	→ -5	→ -6	→ -7
1	G	↓ -1	↘ ↓ → 1						
2	A	↓ -2							
3	T	↓ -3							
4	A	↓ -4							
5	G	↓ -5							
6	A	↓ -6							

Option 1 (↘)

– G
– G

0 + 1 = 1

Option 2 (→)

G –
– G

-1 + -1 = **-2**

Option 3 (↓)

– G
G –

-1 + -1 = **-2**

Option 1 (↘) is the best. Set $S(1,1) = 1$, and keep *only* ↘

Aligning X = GATTACA and Y = GATAGA — EXAMPLE

$Y_j \downarrow$ / $X_i \rightarrow$		0	1	2	3	4	5	6	7
		–	G	A	T	T	A	C	A
0	–	0	→ -1	→ -2	→ -3	→ -4	→ -5	→ -6	→ -7
1	G	↓ -1	↘ 1	↘ ↓ → 0					
2	A	↓ -2							
3	T	↓ -3							
4	A	↓ -4							
5	G	↓ -5							
6	A	↓ -6							

Option 1 (↘)

GA
–G

-1 + 0 = -1

Option 2 (→)

GA
G–

1 + -1 = 0

Option 3 (↓)

GA–
––G

-2 + -1 = -3

Option 2 (→) is the best. Set $S(2,1) = 0$, and keep *only* →

Aligning X = GATTACA and Y = GATAGA **EXAMPLE**

Y_j ↓ \ X_i →		0	1	2	3	4	5	6	7
		–	G	A	T	T	A	C	A
0	–	0	→ -1	→ -2	→ -3	→ -4	→ -5	→ -6	→ -7
1	G	↓ -1	↘ 1	→ 0	↘ ↓ → -1				
2	A	↓ -2							
3	T	↓ -3							
4	A	↓ -4							
5	G	↓ -5							
6	A	↓ -6							

Option 1 (↘)

GAT
--G

-2 + 0 = -2

Option 2 (→)

GAT
G--

0 + -1 = **-1**

Option 3 (↓)

GAT-
---G

-3 + -1 = -4

Option 2 (→) is the best. Set $S(3,1)$ = -1, and keep *only* →

Aligning X = GATTACA and Y = GATAGA — EXAMPLE

	$X_i \rightarrow$ 0	1	2	3	4	5	6	7
$Y_j \downarrow$	–	G	A	T	T	A	C	A
0 –	0	→ -1	→ -2	→ -3	→ -4	→ -5	→ -6	→ -7
1 G	↓ -1	↘ 1	→ 0	→ -1	→ **-2**			
2 A	↓ -2							
3 T	↓ -3							
4 A	↓ -4							
5 G	↓ -5							
6 A	↓ -6							

Option 1 (↘)

T
G

-3 + 0 = **-3**

Option 2 (→)

T
–

-1 + -1 = **-2**

Option 3 (↓)

–
G

-4 + -1 = **-5**

Option 2 (→) is the best. Set $S(4,1) = -2$, and keep *only* →

Aligning X = GATTACA and Y = GATAGA — EXAMPLE

Y_j ↓ \ X_i →		0	1	2	3	4	5	6	7
		–	G	A	T	T	A	C	A
0	–	0	→ -1	→ -2	→ -3	→ -4	→ -5	→ -6	→ -7
1	G	↓ -1	↘ 1	→ 0	→ -1	→ -2	→ **-3**		
2	A	↓ -2							
3	T	↓ -3							
4	A	↓ -4							
5	G	↓ -5							
6	A	↓ -6							

Option 1 (↘)

A
G

-4 + 0 = **-4**

Option 2 (→)

A
–

-2 + -1 = **-3**

Option 3 (↓)

–
G

-5 + -1 = **-6**

Option 2 (→) is the best. Set $S(5,1)$ = -3, and keep *only* →

Aligning X = GATTACA and Y = GATAGA

EXAMPLE

$X_i \rightarrow$		0	1	2	3	4	5	6	7
Y_j ↓		–	G	A	T	T	A	C	A
0	–	0	→ -1	→ -2	→ -3	→ -4	→ -5	→ -6	→ -7
1	G	↓ -1	↘ 1	→ 0	→ -1	→ -2	→ -3	→ **-4**	
2	A	↓ -2							
3	T	↓ -3							
4	A	↓ -4							
5	G	↓ -5							
6	A	↓ -6							

Option 1 (↘)

C
G

-5 + 0 = **-5**

Option 2 (→)

C
–

-3 + -1 = **-4**

Option 3 (↓)

–
G

-6 + -1 = **-7**

Option 2 (→) is the best. Set $\mathcal{S}(6, 1)$ = -4, and keep *only* →

Aligning X = GATTACA and Y = GATAGA

Y_j ↓	X_i →	0	1	2	3	4	5	6	7
		–	G	A	T	T	A	C	A
0	–	0	→ -1	→ -2	→ -3	→ -4	→ -5	→ -6	→ -7
1	G	↓ -1	↘ 1	→ 0	→ -1	→ -2	→ -3	→ -4	→ **-5**
2	A	↓ -2							
3	T	↓ -3							
4	A	↓ -4							
5	G	↓ -5							
6	A	↓ -6							

Option 1 (↘)

A
G

-6 + 0 = **-6**

Option 2 (→)

A
–

-4 + -1 = **-5**

Option 3 (↓)

–
G

-7 + -1 = **-8**

Option 2 (→) is the best. Set $S(7,1)$ = -5, and keep *only* →

Aligning X = GATTACA and Y = GATAGA

EXAMPLE

Y_j ↓ / X_i →		0	1	2	3	4	5	6	7
		–	G	A	T	T	A	C	A
0	–	0	→ -1	→ -2	→ -3	→ -4	→ -5	→ -6	→ -7
1	G	↓ -1	↘ 1	→ 0	→ -1	→ -2	→ -3	→ -4	→ -5
2	A	↓ -2	↓ 0						
3	T	↓ -3							
4	A	↓ -4							
5	G	↓ -5							
6	A	↓ -6							

Option 1 (↘)

```
-G
GA
```

-1 + 0 = **-1**

Option 2 (→)

```
--G
GA-
```

-2 + -1 = **-3**

Option 3 (↓)

```
G-
GA
```

1 + -1 = **0**

Option 3 (↓) is the best. Set $S(1,2) = 0$, and keep *only* ↓

Aligning X = GATTACA and Y = GATAGA

EXAMPLE

		X_i → 0	1	2	3	4	5	6	7
Y_j ↓		–	G	A	T	T	A	C	A
0	–	0	→ -1	→ -2	→ -3	→ -4	→ -5	→ -6	→ -7
1	G	↓ -1	↘ 1	→ 0	→ -1	→ -2	→ -3	→ -4	→ -5
2	A	↓ -2	↓ 0	↘ 2					
3	T	↓ -3							
4	A	↓ -4							
5	G	↓ -5							
6	A	↓ -6							

Option 1 (↘)

A
A

1 + 1 = 2

Option 2 (→)

G–A
GA–

0+ -1 = **-1**

Option 3 (↓)

GA–
G–A

0 + -1 = **-1**

Option 1 (↘) is the best. Set $S(2,2)$ = 2, and keep *only* ↘

Aligning X = GATTACA and Y = GATAGA — EXAMPLE

$Y_j \downarrow$ \ $X_i \rightarrow$		0	1	2	3	4	5	6	7
		–	G	A	T	T	A	C	A
0	–	0	→ -1	→ -2	→ -3	→ -4	→ -5	→ -6	→ -7
1	G	↓ -1	↘ 1	→ 0	→ -1	→ -2	→ -3	→ -4	→ -5
2	A	↓ -2	↓ 0	↘ 2	→ 1				
3	T	↓ -3							
4	A	↓ -4							
5	G	↓ -5							
6	A	↓ -6							

Option 1 (↘)

T
A

0 + 0 = **0**

Option 2 (→)

GAT
GA–

2 + -1 = **1**

Option 3 (↓)

GAT–
G––A

-1 + -1 = **-2**

Option 2 (→) is the best. Set $S(3,2)$ = 1, and keep *only* →

Aligning X = GATTACA and Y = GATAGA

EXAMPLE

Y_j ↓ \ X_i →		0	1	2	3	4	5	6	7
		–	G	A	T	T	A	C	A
0	–	0	→ -1	→ -2	→ -3	→ -4	→ -5	→ -6	→ -7
1	G	↓ -1	↘ 1	→ 0	→ -1	→ -2	→ -3	→ -4	→ -5
2	A	↓ -2	↓ 0	↘ 2	→ 1	→ 0			
3	T	↓ -3							
4	A	↓ -4							
5	G	↓ -5							
6	A	↓ -6							

Option 1 (↘)

T
A

$-1 + 0 = \mathbf{-1}$

Option 2 (→)

T
–

$1 + -1 = 0$

Option 3 (↓)

–
A

$-2 + -1 = \mathbf{-3}$

Option 2 (→) is the best. Set $S(4,2) = 0$, and keep *only* →

Aligning X = GATTACA and Y = GATAGA EXAMPLE

Y_j ↓ \ X_i →		0	1	2	3	4	5	6	7
		–	G	A	T	T	A	C	A
0	–	0	→ -1	→ -2	→ -3	→ -4	→ -5	→ -6	→ -7
1	G	↓ -1	↘ 1	→ 0	→ -1	→ -2	→ -3	→ -4	→ -5
2	A	↓ -2	↓ 0	↘ 2	→ 1	→ 0	↘ → -1		
3	T	↓ -3							
4	A	↓ -4							
5	G	↓ -5							
6	A	↓ -6							

Option 1 (↘)

A
A

$-2 + 1 = -1$

Option 2 (→)

A
–

$0 + -1 = -1$

Option 3 (↓)

–
A

$-3 + -1 = -4$

Options 1 and 2 are both good. Set $S(5,2) = -1$, and keep ↘ and →

Aligning X = GATTACA and Y = GATAGA — EXAMPLE

Y_j ↓ / X_i →		0	1	2	3	4	5	6	7
		–	G	A	T	T	A	C	A
0	–	0	→ -1	→ -2	→ -3	→ -4	→ -5	→ -6	→ -7
1	G	↓ -1	↘ 1	→ 0	→ -1	→ -2	→ -3	→ -4	→ -5
2	A	↓ -2	↓ 0	↘ 2	→ 1	→ 0	↘ → -1	**→ -2**	
3	T	↓ -3							
4	A	↓ -4							
5	G	↓ -5							
6	A	↓ -6							

Option 1 (↘)

C
A

$-3 + 0 = \mathbf{-3}$

Option 2 (→)

C
–

$-1 + -1 = \mathbf{-2}$

Option 3 (↓)

–
A

$-4 + -1 = \mathbf{-5}$

Option 2 is the best. Set $S(6,2) = -2$, and keep *only* →

Aligning X = GATTACA and Y = GATAGA EXAMPLE

Y_j ↓ \ X_i →		0	1	2	3	4	5	6	7
		–	G	A	T	T	A	C	A
0	–	0	→ -1	→ -2	→ -3	→ -4	→ -5	→ -6	→ -7
1	G	↓ -1	↘ 1	→ 0	→ -1	→ -2	→ -3	→ -4	→ -5
2	A	↓ -2	↓ 0	↘ 2	→ 1	→ 0	↘ → -1	→ -2	↘ → **-3**
3	T	↓ -3							
4	A	↓ -4							
5	G	↓ -5							
6	A	↓ -6							

Option 1 (↘)

A
A

-4 + 1 = **-3**

Option 2 (→)

A
–

-2 + -1 = **-3**

Option 3 (↓)

–
A

-5 + -1 = **-6**

Options 1 and 2 are both good. Set $S(7,2)$ = -3, and keep ↘ and →

Aligning X = GATTACA and Y = GATAGA EXAMPLE

		$X_i \rightarrow$ 0	1	2	3	4	5	6	7
$Y_j \downarrow$		–	G	A	T	T	A	C	A
0	–	0	→ -1	→ -2	→ -3	→ -4	→ -5	→ -6	→ -7
1	G	↓ -1	↘ 1	→ 0	→ -1	→ -2	→ -3	→ -4	→ -5
2	A	↓ -2	↓ 0	↘ 2	→ 1	→ 0	↘ → -1	→ -2	↘ → -3
3	T	↓ -3	↓ -1						
4	A	↓ -4							
5	G	↓ -5							
6	A	↓ -6							

Option 1 (↘)

--G
GAT

-2 + 0 = **-2**

Option 2 (→)

---G
GAT-

-3 + -1 = **-4**

Option 3 (↓)

G--
GAT

0 + -1 = **-1**

Option 3 is the best. Set $S(1,3)$ = -1, and keep ↓

Aligning X = GATTACA and Y = GATAGA

EXAMPLE

Y_j ↓ \ X_i →		0	1	2	3	4	5	6	7
		–	G	A	T	T	A	C	A
0	–	0	→ -1	→ -2	→ -3	→ -4	→ -5	→ -6	→ -7
1	G	↓ -1	↘ 1	→ 0	→ -1	→ -2	→ -3	→ -4	→ -5
2	A	↓ -2	↓ 0	↘ 2	→ 1	→ 0	↘ → -1	→ -2	↘ → -3
3	T	↓ -3	↓ -1	↓ 1					
4	A	↓ -4							
5	G	↓ -5							
6	A	↓ -6							

Option 1 (↘)

A
T

0 + 0 = **0**

Option 2 (→)

A
–

-1 + -1 = **-2**

Option 3 (↓)

–
T

2 + -1 = 1

Option 3 is the best. Set $S(2,3)$ = -1, and keep ↓

Aligning X = GATTACA and Y = GATAGA EXAMPLE

Y_j ↓ \ X_i →		0	1	2	3	4	5	6	7
		–	G	A	T	T	A	C	A
0	–	0	→ -1	→ -2	→ -3	→ -4	→ -5	→ -6	→ -7
1	G	↓ -1	↘ 1	→ 0	→ -1	→ -2	→ -3	→ -4	→ -5
2	A	↓ -2	↓ 0	↘ 2	→ 1	→ 0	↘ → -1	→ -2	↘ → -3
3	T	↓ -3	↓ -1	↓ 1	↘ 3				
4	A	↓ -4							
5	G	↓ -5							
6	A	↓ -6							

Option 1 (↘)

T
T

2 + 1 = 3

Option 2 (→)

T
–

-1 + -1 = **-2**

Option 3 (↓)

–
T

1 + -1 = **0**

Option 1 is the best. Set $S(3,3)$ = -1, and keep ↘

Aligning X = GATTACA and Y = GATAGA EXAMPLE

Y_j ↓ \ X_i →	0 –	1 G	2 A	3 T	4 T	5 A	6 C	7 A
0 –	0	→ -1	→ -2	→ -3	→ -4	→ -5	→ -6	→ -7
1 G	↓ -1	↘ 1	→ 0	→ -1	→ -2	→ -3	→ -4	→ -5
2 A	↓ -2	↓ 0	↘ 2	→ 1	→ 0	↘ → -1	→ -2	↘ → -3
3 T	↓ -3	↓ -1	↓ 1	↘ 3	↘ → **2**			
4 A	↓ -4							
5 G	↓ -5							
6 A	↓ -6							

Option 1 (↘)

T
T

1 + 1 = **2**

Option 2 (→)

T
–

3 + -1 = **2**

Option 3 (↓)

–
T

0 + -1 = **-1**

Options 1 & 2 are both good. Set $S(4,3) = 2$, keep both ↘ and →

Aligning X = GATTACA and Y = GATAGA

EXAMPLE

Y_j ↓ / X_i →		0	1	2	3	4	5	6	7
		–	G	A	T	T	A	C	A
0	–	0	→ -1	→ -2	→ -3	→ -4	→ -5	→ -6	→ -7
1	G	↓ -1	↘ 1	→ 0	→ -1	→ -2	→ -3	→ -4	→ -5
2	A	↓ -2	↓ 0	↘ 2	→ 1	→ 0	↘ → -1	→ -2	↘ → -3
3	T	↓ -3	↓ -1	↓ 1	↘ 3	↘ → 2	→ 1		
4	A	↓ -4							
5	G	↓ -5							
6	A	↓ -6							

Option 1 (↘)

A
T

0 + 0 = **0**

Option 2 (→)

A
–

2 + -1 = **1**

Option 3 (↓)

–
T

-1 + -1 = **-2**

Option 2 is the best. Set $S(5,3)$ = 1, keep *only* →

Aligning X = GATTACA and Y = GATAGA — EXAMPLE

Y_j ↓ / X_i →		0	1	2	3	4	5	6	7
		–	G	A	T	T	A	C	A
0	–	0	→ -1	→ -2	→ -3	→ -4	→ -5	→ -6	→ -7
1	G	↓ -1	↘ 1	→ 0	→ -1	→ -2	→ -3	→ -4	→ -5
2	A	↓ -2	↓ 0	↘ 2	→ 1	→ 0	↘ → -1	→ -2	↘ → -3
3	T	↓ -3	↓ -1	↓ 1	↘ 3	↘ → 2	→ 1	→ 0	
4	A	↓ -4							
5	G	↓ -5							
6	A	↓ -6							

Option 1 (↘)

C
T

$-1 + 0 = \mathbf{-1}$

Option 2 (→)

C
–

$1 + -1 = \mathbf{0}$

Option 3 (↓)

–
T

$-2 + -1 = \mathbf{-3}$

Option 2 is the best. Set $S(6,3) = 0$, keep *only* →

Aligning X = GATTACA and Y = GATAGA EXAMPLE

Y_j ↓ / X_i →	0	1	2	3	4	5	6	7
	–	G	A	T	T	A	C	A
0 –	0	→ -1	→ -2	→ -3	→ -4	→ -5	→ -6	→ -7
1 G	↓ -1	↘ 1	→ 0	→ -1	→ -2	→ -3	→ -4	→ -5
2 A	↓ -2	↓ 0	↘ 2	→ 1	→ 0	↘ → -1	→ -2	↘ → -3
3 T	↓ -3	↓ -1	↓ 1	↘ 3	↘ → 2	→ 1	→ 0	→ -1
4 A	↓ -4							
5 G	↓ -5							
6 A	↓ -6							

Option 1 (↘)

A
T

-2 + 0 = **-2**

Option 2 (→)

A
–

0 + -1 = **-1**

Option 3 (↓)

–
T

-3 + -1 = **-4**

Option 2 is the best. Set $S(7,3)$ = 0, keep *only* →

Aligning X = GATTACA and Y = GATAGA EXAMPLE

Y_j ↓ / X_i →		0	1	2	3	4	5	6	7
		–	G	A	T	T	A	C	A
0	–	0	→ -1	→ -2	→ -3	→ -4	→ -5	→ -6	→ -7
1	G	↓ -1	↘ 1	→ 0	→ -1	→ -2	→ -3	→ -4	→ -5
2	A	↓ -2	↓ 0	↘ 2	→ 1	→ 0	↘ → -1	→ -2	↘ → -3
3	T	↓ -3	↓ -1	↓ 1	↘ 3	↘ → 2	→ 1	→ 0	→ -1
4	A	↓ -4	↓ **-2**						
5	G	↓ -5							
6	A	↓ -6							

Option 1 (↘)

G
A

-3 + 0 = **-3**

Option 2 (→)

G
–

-4 + -1 = **-5**

Option 3 (↓)

–
A

-1 + -1 = **-2**

Option 3 is the best. Set $S(1,4)$ = -2, keep *only* ↓

Aligning X = GATTACA and Y = GATAGA EXAMPLE

	X_i →	0	1	2	3	4	5	6	7
Y_j ↓		–	G	A	T	T	A	C	A
0	–	0	→ -1	→ -2	→ -3	→ -4	→ -5	→ -6	→ -7
1	G	↓ -1	↘ 1	→ 0	→ -1	→ -2	→ -3	→ -4	→ -5
2	A	↓ -2	↓ 0	↘ 2	→ 1	→ 0	↘ → -1	→ -2	↘ → -3
3	T	↓ -3	↓ -1	↓ 1	↘ 3	↘ → 2	→ 1	→ 0	→ -1
4	A	↓ -4	↓ -2	↘ ↓ 0					
5	G	↓ -5							
6	A	↓ -6							

Option 1 (↘)

A
A

-1 + 1 = 0

Option 2 (→)

A
–

-2 + -1 = -3

Option 3 (↓)

–
A

1 + -1 = 0

Options 1 & 3 are both good. Set $S(2,4) = 0$, keep both ↘ and →

Aligning X = GATTACA and Y = GATAGA   EXAMPLE

	$X_i \rightarrow$	0	1	2	3	4	5	6	7
$Y_j \downarrow$		–	G	A	T	T	A	C	A
0	–	0	→ -1	→ -2	→ -3	→ -4	→ -5	→ -6	→ -7
1	G	↓ -1	↘ 1	→ 0	→ -1	→ -2	→ -3	→ -4	→ -5
2	A	↓ -2	↓ 0	↘ 2	→ 1	→ 0	↘ → -1	→ -2	↘ → -3
3	T	↓ -3	↓ -1	↓ 1	↘ 3	↘ → 2	→ 1	→ 0	→ -1
4	A	↓ -4	↓ -2	↘ ↓ 0	↓ 2				
5	G	↓ -5							
6	A	↓ -6							

Option 1 (↘)

T
A

0 + 0 = **0**

Option 2 (→)

T
–

0 + -1 = **-1**

Option 3 (↓)

–
A

3 + -1 = **2**

Option 3 is the best. Set $S(3,4) = 2$, keep *only* ↓

Aligning X = GATTACA and Y = GATAGA — EXAMPLE

Y_j ↓ \ X_i →		0	1	2	3	4	5	6	7
		–	G	A	T	T	A	C	A
0	–	0	→ -1	→ -2	→ -3	→ -4	→ -5	→ -6	→ -7
1	G	↓ -1	↘ 1	→ 0	→ -1	→ -2	→ -3	→ -4	→ -5
2	A	↓ -2	↓ 0	↘ 2	→ 1	→ 0	↘ → -1	→ -2	↘ → -3
3	T	↓ -3	↓ -1	↓ 1	↘ 3	↘ → 2	→ 1	→ 0	→ -1
4	A	↓ -4	↓ -2	↘ ↓ 0	↓ 2	↘ **3**			
5	G	↓ -5							
6	A	↓ -6							

Option 1 (↘)

T
A
$3 + 0 = \mathbf{3}$

Option 2 (→)

T
–
$2 + -1 = 1$

Option 3 (↓)

–
A
$2 + -1 = 1$

Option 1 is the best. Set $S(4,4) = 3$, keep *only* ↘

Aligning X = GATTACA and Y = GATAGA — EXAMPLE

Y_j ↓ \ X_i →		0	1	2	3	4	5	6	7
		–	G	A	T	T	A	C	A
0	–	0	→ -1	→ -2	→ -3	→ -4	→ -5	→ -6	→ -7
1	G	↓ -1	↘ 1	→ 0	→ -1	→ -2	→ -3	→ -4	→ -5
2	A	↓ -2	↓ 0	↘ 2	→ 1	→ 0	↘ → -1	→ -2	↘ → -3
3	T	↓ -3	↓ -1	↓ 1	↘ 3	↘ → 2	→ 1	→ 0	→ -1
4	A	↓ -4	↓ -2	↘ ↓ 0	↓ 2	↘ 3	↘ 3		
5	G	↓ -5							
6	A	↓ -6							

Option 1 (↘)

A
A

2 + 1 = 3

Option 2 (→)

A
–

3 + -1 = 2

Option 3 (↓)

–
A

1 + -1 = 0

Option 1 is the best. Set $S(5,4) = 3$, keep *only* ↘

Aligning X = GATTACA and Y = GATAGA

EXAMPLE

Y_j ↓ \ X_i →	0	1	2	3	4	5	6	7
	–	G	A	T	T	A	C	A
0 –	0	→ -1	→ -2	→ -3	→ -4	→ -5	→ -6	→ -7
1 G	↓ -1	↘ 1	→ 0	→ -1	→ -2	→ -3	→ -4	→ -5
2 A	↓ -2	↓ 0	↘ 2	→ 1	→ 0	↘ → -1	→ -2	↘ → -3
3 T	↓ -3	↓ -1	↓ 1	↘ 3	↘ → 2	→ 1	→ 0	→ -1
4 A	↓ -4	↓ -2	↘ ↓ 0	↓ 2	↘ 3	↘ 3	→ 2	
5 G	↓ -5							
6 A	↓ -6							

Option 1 (↘)

C
A $\quad 1 + 0 = 1$

Option 2 (→)

C
– $\quad 3 + -1 = 2$

Option 3 (↓)

–
A $\quad 0 + -1 = -1$

Option 2 is the best. Set $S(6,4) = 2$, keep *only* →

Aligning X = GATTACA and Y = GATAGA

EXAMPLE

$Y_j \downarrow$ / $X_i \rightarrow$		0	1	2	3	4	5	6	7
		–	G	A	T	T	A	C	A
0	–	0	→ -1	→ -2	→ -3	→ -4	→ -5	→ -6	→ -7
1	G	↓ -1	↘ 1	→ 0	→ -1	→ -2	→ -3	→ -4	→ -5
2	A	↓ -2	↓ 0	↘ 2	→ 1	→ 0	↘ → -1	→ -2	↘ → -3
3	T	↓ -3	↓ -1	↓ 1	↘ 3	↘ → 2	→ 1	→ 0	→ -1
4	A	↓ -4	↓ -2	↘ ↓ 0	↓ 2	↘ 3	↘ 3	→ 2	↘ → 1
5	G	↓ -5							
6	A	↓ -6							

Option 1 (↘)

A
A

0 + 1 = 1

Option 2 (→)

A
–

2 + -1 = 1

Option 3 (↓)

–
A

-1 + -1 = -2

Options 1 & 2 are both good. Set $S(7,4) = 1$, keep both ↘ and →

Aligning X = GATTACA and Y = GATAGA

EXAMPLE

Y_j ↓ / X_i →		0	1	2	3	4	5	6	7
		–	G	A	T	T	A	C	A
0	–	0	→ -1	→ -2	→ -3	→ -4	→ -5	→ -6	→ -7
1	G	↓ -1	↘ 1	→ 0	→ -1	→ -2	→ -3	→ -4	→ -5
2	A	↓ -2	↓ 0	↘ 2	→ 1	→ 0	↘ → -1	→ -2	↘ → -3
3	T	↓ -3	↓ -1	↓ 1	↘ 3	↘ → 2	→ 1	→ 0	→ -1
4	A	↓ -4	↓ -2	↘ ↓ 0	↓ 2	↘ 3	↘ 3	→ 2	↘ → 1
5	G	↓ -5	↘ ↓ -3						
6	A	↓ -6							

Option 1 (↘)

G
G

-4 + 1 = -3

Option 2 (→)

G
–

-5 + -1 = -6

Option 3 (↓)

–
G

-2 + -1 = -3

Options 1 & 3 are both good. Set $S(1,5) = -3$, keep both ↘ and ↓

Aligning X = GATTACA and Y = GATAGA — EXAMPLE

Y_j ↓ \ X_i →		0	1	2	3	4	5	6	7
		–	G	A	T	T	A	C	A
0	–	0	→ -1	→ -2	→ -3	→ -4	→ -5	→ -6	→ -7
1	G	↓ -1	↘ 1	→ 0	→ -1	→ -2	→ -3	→ -4	→ -5
2	A	↓ -2	↓ 0	↘ 2	→ 1	→ 0	↘ → -1	→ -2	↘ → -3
3	T	↓ -3	↓ -1	↓ 1	↘ 3	↘ → 2	→ 1	→ 0	→ -1
4	A	↓ -4	↓ -2	↘ ↓ 0	↓ 2	↘ 3	↘ 3	→ 2	↘ → 1
5	G	↓ -5	↘ ↓ -3	↓ -1					
6	A	↓ -6							

Option 1 (↘)

A
G
-2 + 0 = **-2**

Option 2 (→)

A
–
-3 + -1 = **-4**

Option 3 (↓)

–
G
0 + -1 = **-1**

Option 3 is the best. Set $S(2,5)$ = -1, keep *only* ↓

Aligning X = GATTACA and Y = GATAGA

EXAMPLE

$Y_j \downarrow$ / $X_i \rightarrow$		0	1	2	3	4	5	6	7
		–	G	A	T	T	A	C	A
0	–	0	→ -1	→ -2	→ -3	→ -4	→ -5	→ -6	→ -7
1	G	↓ -1	↘ 1	→ 0	→ -1	→ -2	→ -3	→ -4	→ -5
2	A	↓ -2	↓ 0	↘ 2	→ 1	→ 0	↘ → -1	→ -2	↘ → -3
3	T	↓ -3	↓ -1	↓ 1	↘ 3	↘ → 2	→ 1	→ 0	→ -1
4	A	↓ -4	↓ -2	↘ ↓ 0	↓ 2	↘ 3	↘ 3	→ 2	↘ → 1
5	G	↓ -5	↘ ↓ -3	↓ -1	↓ 1				
6	A	↓ -6							

Option 1 (↘)

T
G

$0 + 0 = \mathbf{0}$

Option 2 (→)

T
–

$-1 + -1 = \mathbf{-2}$

Option 3 (↓)

–
G

$2 + -1 = 1$

Option 3 is the best. Set $S(3,5) = 1$, keep *only* ↓

Aligning X = GATTACA and Y = GATAGA

EXAMPLE

Y_j ↓ \ X_i →	0	1	2	3	4	5	6	7
	–	G	A	T	T	A	C	A
0 –	0	→ -1	→ -2	→ -3	→ -4	→ -5	→ -6	→ -7
1 G	↓ -1	↘ 1	→ 0	→ -1	→ -2	→ -3	→ -4	→ -5
2 A	↓ -2	↓ 0	↘ 2	→ 1	→ 0	↘ → -1	→ -2	↘ → -3
3 T	↓ -3	↓ -1	↓ 1	↘ 3	↘ → 2	→ 1	→ 0	→ -1
4 A	↓ -4	↓ -2	↘ ↓ 0	↓ 2	↘ 3	↘ 3	→ 2	↘ → 1
5 G	↓ -5	↘ ↓ -3	↓ -1	↓ 1	↘ ↓ 2			
6 A	↓ -6							

Option 1 (↘)

T
G

$2 + 0 = 2$

Option 2 (→)

T
–

$1 + -1 = 0$

Option 3 (↓)

–
G

$3 + -1 = 2$

Options 1 and 3 are both good. Set $S(4,5) = 2$, keep both ↘ and ↓

Aligning X = GATTACA and Y = GATAGA

EXAMPLE

$Y_j \downarrow$ / $X_i \rightarrow$		0	1	2	3	4	5	6	7
		–	G	A	T	T	A	C	A
0	–	0	→ -1	→ -2	→ -3	→ -4	→ -5	→ -6	→ -7
1	G	↓ -1	↘ 1	→ 0	→ -1	→ -2	→ -3	→ -4	→ -5
2	A	↓ -2	↓ 0	↘ 2	→ 1	→ 0	↘ → -1	→ -2	↘ → -3
3	T	↓ -3	↓ -1	↓ 1	↘ 3	↘ → 2	→ 1	→ 0	→ -1
4	A	↓ -4	↓ -2	↘ ↓ 0	↓ 2	↘ 3	↘ 3	→ 2	↘ → 1
5	G	↓ -5	↘ ↓ -3	↓ -1	↓ 1	↘ ↓ 2	↘ 3		
6	A	↓ -6							

Option 1 (↘)

A
G

3 + 0 = 3

Option 2 (→)

A
–

2 + -1 = 1

Option 3 (↓)

–
G

3 + -1 = 2

Option 1 is the best. Set $S(5,5)$ = 3, keep *only* ↘

Aligning X = GATTACA and Y = GATAGA

EXAMPLE

$Y_j \downarrow$ / $X_i \rightarrow$		0	1	2	3	4	5	6	7
		–	G	A	T	T	A	C	A
0	–	0	→ -1	→ -2	→ -3	→ -4	→ -5	→ -6	→ -7
1	G	↓ -1	↘ 1	→ 0	→ -1	→ -2	→ -3	→ -4	→ -5
2	A	↓ -2	↓ 0	↘ 2	→ 1	→ 0	↘ → -1	→ -2	↘ → -3
3	T	↓ -3	↓ -1	↓ 1	↘ 3	↘ → 2	→ 1	→ 0	→ -1
4	A	↓ -4	↓ -2	↘ ↓ 0	↓ 2	↘ 3	↘ 3	→ 2	↘ → 1
5	G	↓ -5	↘ ↓ -3	↓ -1	↓ 1	↘ ↓ 2	↘ 3	↘ 3	
6	A	↓ -6							

Option 1 (↘)

C
G

3 + 0 = 3

Option 2 (→)

C
–

3 + -1 = 2

Option 3 (↓)

–
G

2 + -1 = 1

Option 1 is the best. Set $S(6,5)$ = 3, keep *only* ↘

Aligning X = GATTACA and Y = GATAGA — EXAMPLE

Y_j ↓ \ X_i →		0	1	2	3	4	5	6	7
		–	G	A	T	T	A	C	A
0	–	0	→ -1	→ -2	→ -3	→ -4	→ -5	→ -6	→ -7
1	G	↓ -1	↘ 1	→ 0	→ -1	→ -2	→ -3	→ -4	→ -5
2	A	↓ -2	↓ 0	↘ 2	→ 1	→ 0	↘ → -1	→ -2	↘ → -3
3	T	↓ -3	↓ -1	↓ 1	↘ 3	↘ → 2	→ 1	→ 0	→ -1
4	A	↓ -4	↓ -2	↘ ↓ 0	↓ 2	↘ 3	↘ 3	→ 2	↘ → 1
5	G	↓ -5	↘ ↓ -3	↓ -1	↓ 1	↘ ↓ 2	↘ 3	↘ 3	↘ → 2
6	A	↓ -6							

Option 1 (↘)

A
G

2 + 0 = 2

Option 2 (→)

A
–

3 + -1 = 2

Option 3 (↓)

–
G

1 + -1 = 0

Options 1 and 2 are both good. Set $S(7,5) = 2$, keep both ↘ and →

Aligning X = GATTACA and Y = GATAGA

EXAMPLE

Y_j ↓ \ X_i →		0	1	2	3	4	5	6	7
		–	G	A	T	T	A	C	A
0	–	0	→ -1	→ -2	→ -3	→ -4	→ -5	→ -6	→ -7
1	G	↓ -1	↘ 1	→ 0	→ -1	→ -2	→ -3	→ -4	→ -5
2	A	↓ -2	↓ 0	↘ 2	→ 1	→ 0	↘ → -1	→ -2	↘ → -3
3	T	↓ -3	↓ -1	↓ 1	↘ 3	↘ → 2	→ 1	→ 0	→ -1
4	A	↓ -4	↓ -2	↘ ↓ 0	↓ 2	↘ 3	↘ 3	→ 2	↘ → 1
5	G	↓ -5	↘ ↓ -3	↓ -1	↓ 1	↘ ↓ 2	↘ 3	↘ 3	↘ → 2
6	A	↓ -6	↓ **-4**						

Option 1 (↘)

G
A
-5 + 0 = **-5**

Option 2 (→)

G
–
-6 + -1 = **-7**

Option 3 (↓)

–
A
-3 + -1 = **-4**

Option 3 is the best. Set $S(1,6)$ = -4, keep *only* ↓

Aligning X = GATTACA and Y = GATAGA — EXAMPLE

Y_j ↓ / X_i →		0 –	1 G	2 A	3 T	4 T	5 A	6 C	7 A
0	–	0	→ -1	→ -2	→ -3	→ -4	→ -5	→ -6	→ -7
1	G	↓ -1	↘ 1	→ 0	→ -1	→ -2	→ -3	→ -4	→ -5
2	A	↓ -2	↓ 0	↘ 2	→ 1	→ 0	↘ → -1	→ -2	↘ → -3
3	T	↓ -3	↓ -1	↓ 1	↘ 3	↘ → 2	→ 1	→ 0	→ -1
4	A	↓ -4	↓ -2	↘ ↓ 0	↓ 2	↘ 3	↘ 3	→ 2	↘ → 1
5	G	↓ -5	↘ ↓ -3	↓ -1	↓ 1	↘ ↓ 2	↘ 3	↘ 3	↘ → 2
6	A	↓ -6	↓ -4	↘ ↓ -2					

Option 1 (↘)

A
A

$-3 + 1 = -2$

Option 2 (→)

A
–

$-4 + -1 = -5$

Option 3 (↓)

–
A

$-1 + -1 = -2$

Options 1 and 3 are both good. Set $\mathcal{S}(2,6) = -2$, keep both ↘ and ↓

Aligning X = GATTACA and Y = GATAGA

EXAMPLE

$X_i \rightarrow$		0	1	2	3	4	5	6	7
$Y_j \downarrow$		–	G	A	T	T	A	C	A
0	–	0	→ -1	→ -2	→ -3	→ -4	→ -5	→ -6	→ -7
1	G	↓ -1	↘ 1	→ 0	→ -1	→ -2	→ -3	→ -4	→ -5
2	A	↓ -2	↓ 0	↘ 2	→ 1	→ 0	↘ → -1	→ -2	↘ → -3
3	T	↓ -3	↓ -1	↓ 1	↘ 3	↘ → 2	→ 1	→ 0	→ -1
4	A	↓ -4	↓ -2	↘ ↓ 0	↓ 2	↘ 3	↘ 3	→ 2	↘ → 1
5	G	↓ -5	↘ ↓ -3	↓ -1	↓ 1	↘ ↓ 2	↘ 3	↘ 3	↘ → 2
6	A	↓ -6	↓ -4	↘ ↓ -2	↓ 0				

Option 1 (↘)

T
A

-1 + 0 = **-1**

Option 2 (→)

T
–

-2 + -1 = **-3**

Option 3 (↓)

–
A

1 + -1 = **0**

Option 3 is the best. Set $S(3,6)$ = 0, keep *only* ↓

Aligning X = GATTACA and Y = GATAGA — EXAMPLE

Y_j ↓ / X_i →	0	1	2	3	4	5	6	7
	–	G	A	T	T	A	C	A
0 –	0	→ -1	→ -2	→ -3	→ -4	→ -5	→ -6	→ -7
1 G	↓ -1	↘ 1	→ 0	→ -1	→ -2	→ -3	→ -4	→ -5
2 A	↓ -2	↓ 0	↘ 2	→ 1	→ 0	↘ → -1	→ -2	↘ → -3
3 T	↓ -3	↓ -1	↓ 1	↘ 3	↘ → 2	→ 1	→ 0	→ -1
4 A	↓ -4	↓ -2	↘ ↓ 0	↓ 2	↘ 3	↘ 3	→ 2	↘ → 1
5 G	↓ -5	↘ ↓ -3	↓ -1	↓ 1	↘ ↓ 2	↘ 3	↘ 3	↘ → 2
6 A	↓ -6	↓ -4	↘ ↓ -2	↓ 0	↘ ↓ 1			

Option 1 (↘)

T
A

1 + 0 = 1

Option 2 (→)

T
–

0 + -1 = -1

Option 3 (↓)

–
A

2 + -1 = 1

Options 1 and 3 are both good. Set $S(4,6) = 1$, keep both ↘ and ↓

Aligning X = GATTACA and Y = GATAGA

EXAMPLE

Y_j ↓ \ X_i →	0 –	1 G	2 A	3 T	4 T	5 A	6 C	7 A
0 –	0	→ -1	→ -2	→ -3	→ -4	→ -5	→ -6	→ -7
1 G	↓ -1	↘ 1	→ 0	→ -1	→ -2	→ -3	→ -4	→ -5
2 A	↓ -2	↓ 0	↘ 2	→ 1	→ 0	↘ → -1	→ -2	↘ → -3
3 T	↓ -3	↓ -1	↓ 1	↘ 3	↘ → 2	→ 1	→ 0	→ -1
4 A	↓ -4	↓ -2	↘ ↓ 0	↓ 2	↘ 3	↘ 3	→ 2	↘ → 1
5 G	↓ -5	↘ ↓ -3	↓ -1	↓ 1	↘ ↓ 2	↘ 3	↘ 3	↘ → 2
6 A	↓ -6	↓ -4	↘ ↓ -2	↓ 0	↘ ↓ 1	↘ 3		

Option 1 (↘)

A
A

2 + 1 = 3

Option 2 (→)

A
–

1 + -1 = 0

Option 3 (↓)

–
A

3 + -1 = 2

Option 1 is the best. Set $S(5,6) = 3$, keep *only* ↘

Aligning X = GATTACA and Y = GATAGA — EXAMPLE

Y_j ↓ \ X_i →	0 –	1 G	2 A	3 T	4 T	5 A	6 C	7 A
0 –	0	→ -1	→ -2	→ -3	→ -4	→ -5	→ -6	→ -7
1 G	↓ -1	↘ 1	→ 0	→ -1	→ -2	→ -3	→ -4	→ -5
2 A	↓ -2	↓ 0	↘ 2	→ 1	→ 0	↘ → -1	→ -2	↘ → -3
3 T	↓ -3	↓ -1	↓ 1	↘ 3	↘ → 2	→ 1	→ 0	→ -1
4 A	↓ -4	↓ -2	↘ ↓ 0	↓ 2	↘ 3	↘ 3	→ 2	↘ → 1
5 G	↓ -5	↘ ↓ -3	↓ -1	↓ 1	↘ ↓ 2	↘ 3	↘ 3	↘ → 2
6 A	↓ -6	↓ -4	↘ ↓ -2	↓ 0	↘ ↓ 1	↘ 3	↘ 3	↘ 4

Option 1 (↘)

A
A

3 + 1 = 4

Option 2 (→)

A
–

3 + -1 = 2

Option 3 (↓)

–
A

2 + -1 = 1

Option 1 is the best. Set $S(7,6)$ = 4, keep *only* ↘

Aligning X = GATTACA and Y = GATAGA

$Y_j \downarrow$ \ $X_i \rightarrow$		0	1	2	3	4	5	6	7
		–	G	A	T	T	A	C	A
0	–	0	→ -1	→ -2	→ -3	→ -4	→ -5	→ -6	→ -7
1	G	↓ -1	↘ 1	→ 0	→ -1	→ -2	→ -3	→ -4	→ -5
2	A	↓ -2	↓ 0	↘ 2	→ 1	→ 0	↘ → -1	→ -2	↘ → -3
3	T	↓ -3	↓ -1	↓ 1	↘ 3	↘ → 2	→ 1	→ 0	→ -1
4	A	↓ -4	↓ -2	↘ ↓ 0	↓ 2	↘ 3	↘ 3	→ 2	↘ → 1
5	G	↓ -5	↘ ↓ -3	↓ -1	↓ 1	↘ ↓ 2	↘ 3	↘ 3	↘ → 2
6	A	↓ -6	↓ -4	↘ ↓ -2	↓ 0	↘ ↓ 1	↘ 3	↘ 3	↘ 4

Here is the completed table!

How do we know what the best alignment is?

Aligning X = GATTACA and Y = GATAGA — EXAMPLE

$X_i \rightarrow$		0	1	2	3	4	5	6	7
$Y_j \downarrow$		–	G	A	T	T	A	C	A
0	–	0	→ -1	→ -2	→ -3	→ -4	→ -5	→ -6	→ -7
1	G	↓ -1	↖ 1	→ 0	→ -1	→ -2	→ -3	→ -4	→ -5
2	A	↓ -2	↓ 0	↖ 2	→ 1	→ 0	↘ → -1	→ -2	↘ → -3
3	T	↓ -3	↓ -1	↓ 1	↖ 3	↘ ← 2	→ 1	→ 0	→ -1
4	A	↓ -4	↓ -2	↘ ↓ 0	↓ 2	↘ 3	↖ 3	→ 2	↘ → 1
5	G	↓ -5	↘ ↓ -3	↓ -1	↓ 1	↘ ↓ 2	↘ 3	↖ 3	↘ → 2
6	A	↓ -6	↓ -4	↘ ↓ -2	↓ 0	↘ ↓ 1	↘ 3	↘ 3	↖ 4

```
G A T T A C A
| | |   |   |
G A T – A G A
```

To find the best **global** alignment we trace back from the **bottom right** corner!

Follow the arrows!

(Do not simply go back to the next largest number in the table)

Aligning X = GATTACA and Y = GATAGA — EXAMPLE

Y_j ↓ \ X_i →	0 –	1 G	2 A	3 T	4 T	5 A	6 C	7 A
0 –	0	→ -1	→ -2	→ -3	→ -4	→ -5	→ -6	→ -7
1 G	↓ -1	↖ 1	→ 0	→ -1	→ -2	→ -3	→ -4	→ -5
2 A	↓ -2	↓ 0	↖ 2	← 1	→ 0	↘ → -1	→ -2	↘ → -3
3 T	↓ -3	↓ -1	↓ 1	↘ 3	↖ → 2	→ 1	→ 0	→ -1
4 A	↓ -4	↓ -2	↘ ↓ 0	↓ 2	↘ 3	↖ 3	→ 2	↘ → 1
5 G	↓ -5	↘ ↓ -3	↓ -1	↓ 1	↘ ↓ 2	↘ 3	↖ 3	↘ → 2
6 A	↓ -6	↓ -4	↘ ↓ -2	↓ 0	↓ 1	↘ 3	↘ 3	↖ 4

G	A	T	T	A	C	A
|	|		|	|		|
G	A	–	T	A	G	A

To find the best **global** alignment we trace back from the **bottom right** corner!

Follow the **arrows!**

There may be more than one optimal alignment (*ties*)!

LOCAL ALIGNMENT

Global alignment

	X_i	G_1	A_2	T_3	T_4	A_5	C_6	A_7
Y_j	**0**	-1	-2	-3	-4	-5	-6	-7
G_1	-1	**1**	0	-1	-2	-3	-4	-5
A_2	-2	0	**2**	1	0	-1	-2	-3
T_3	-3	-1	1	**3**	**2**	1	0	-1
A_4	-4	-2	0	2	3	**3**	2	1
G_5	-5	-3	-1	1	2	3	**3**	2
A_6	-6	-4	-2	0	1	3	3	**4**

Local alignment

	X_i	G_1	A_2	T_3	T_4	A_5	C_6	A_7
Y_j	**0**	0	0	0	0	0	0	0
G_1	0	**1**	0	0	0	0	0	0
A_2	0	0	**2**	0	0	1	0	1
T_3	0	0	0	**3**	1	0	0	0
A_4	0	0	1	1	2	2	0	1
G_5	0	1	0	0	0	1	1	0
A_6	0	0	2	0	0	1	0	2

```
G A T T A C A
| | |   |   |
G A T – A G A
```

Note the local alignment is shorter!

```
G A T
| | |
G A T
```

LOCAL ALIGNMENT

Global alignment

	X_i	G_1	A_2	T_3	T_4	A_5	C_6	A_7
Y_j	0	-1	-2	-3	-4	-5	-6	-7
G_1	-1	1	0	-1	-2	-3	-4	-5
A_2	-2	0	2	1	0	-1	-2	-3
T_3	-3	-1	1	3	2	1	0	-1
A_4	-4	-2	0	2	3	3	2	1
G_5	-5	-3	-1	1	2	3	3	2
A_6	-6	-4	-2	0	1	3	3	4

```
GATTACA
||| | |
GAT-AGA
```

Local alignment

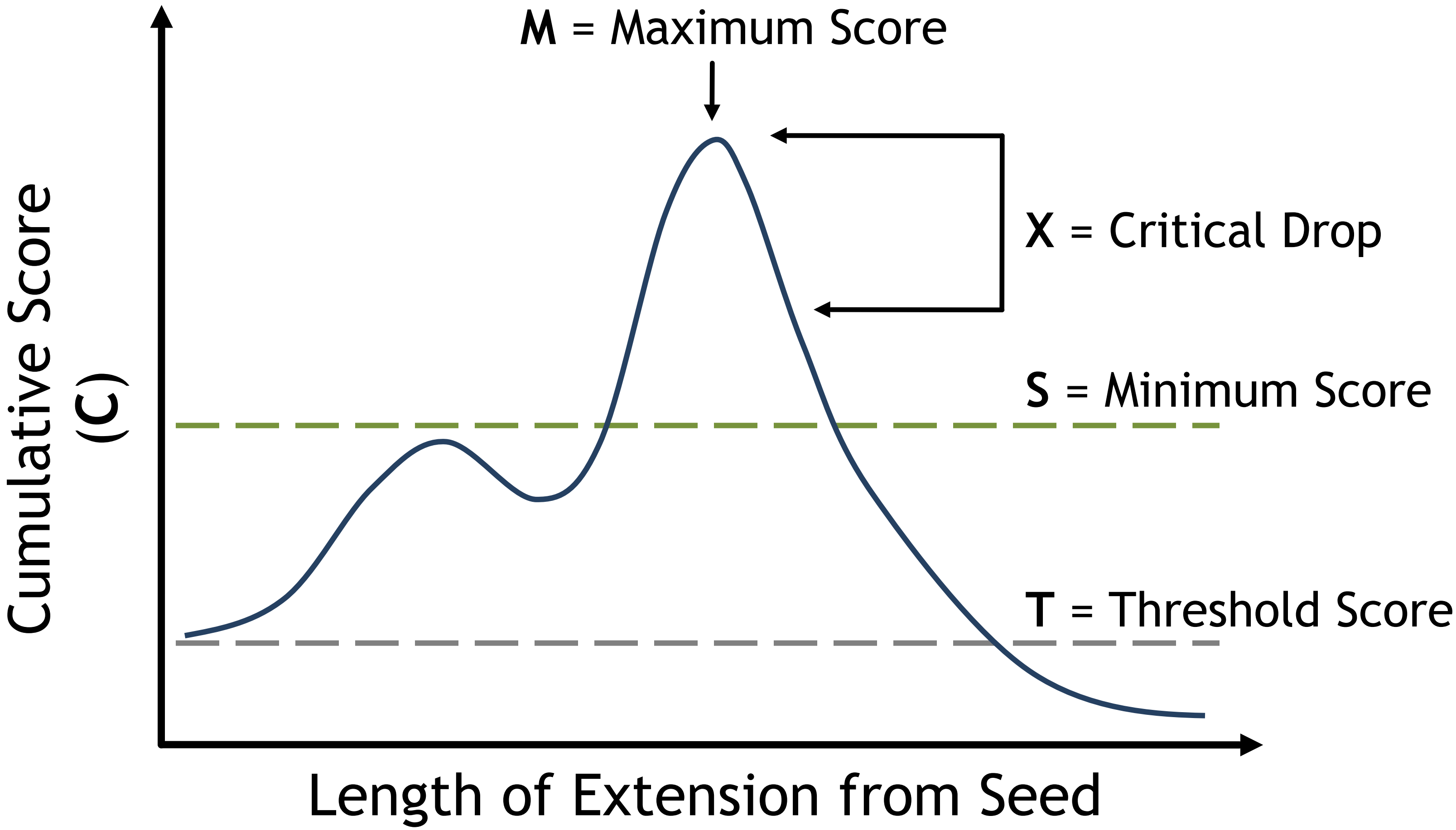
	X_i	G_1	A_2	T_3	T_4	A_5	C_6	A_7
Y_j	0	0	0	0	0	0	0	0
G_1	0	1	0	0	0	0	0	0
A_2	0	0	2	0	0	1	0	1
T_3	0	0	0	3	1	0	0	0
A_4	0	0	1	1	2	2	0	1
G_5	0	1	0	0	0	1	1	0
A_6	0	0	2	0	0	1	0	2

```
GAT
|||
GAT
```

On the board: BLAST extension

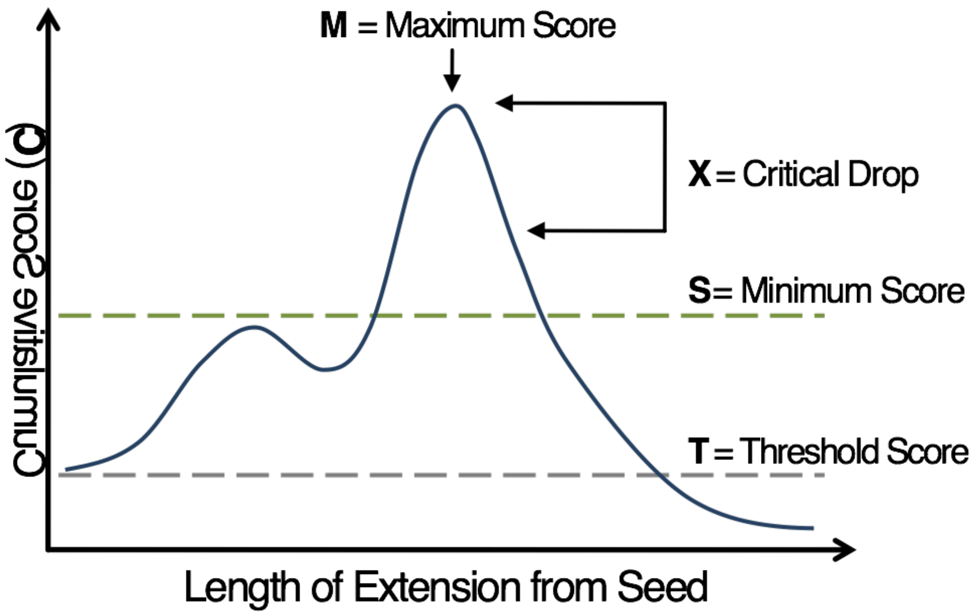
(must answer)

Seed Extension



A good seed has score **T**. *Extend the seed, updating the cumulative score* **C** *and max score* **M**. *Stop when* **C** *falls* **X** *or more units below* **M**. *If* **M** > **S**, *then this extension is a high-scoring segment pair (HSP).*

Seed Extension



Example:

T = 10
S = 20
X = 8

M =
C =

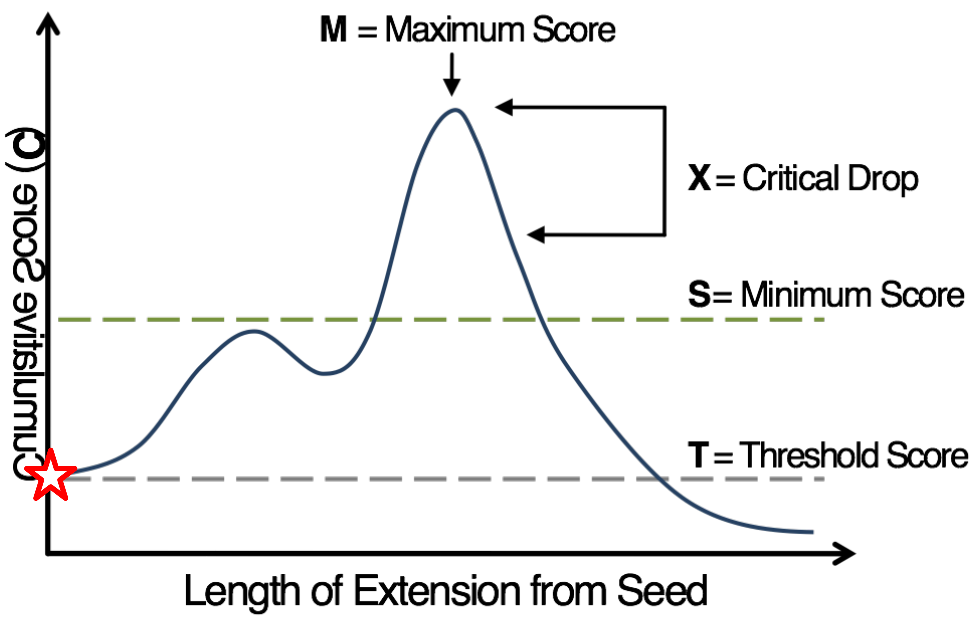
Query sequence

M R L M G A L E D V A L N A I

C Q D M C S L E E I T L W C Q

Database sequence

Seed Extension



Example:

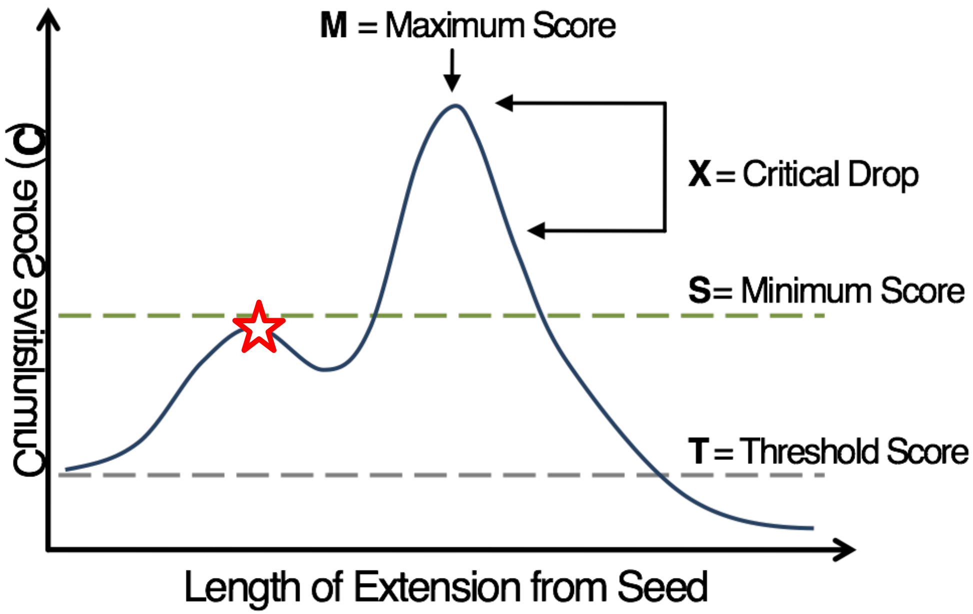
T = 10
S = 20
X = 8

M = 11
C = 11

M	R	L	M	G	A	**L**	**E**	**D**	V	A	L	N	A	I
C	Q	D	M	C	S	**L**	**E**	**E**	I	T	L	W	C	Q
						+4	+5	+2						

This is a good seed (>T), even though it isn't an exact match.

Seed Extension



Example:

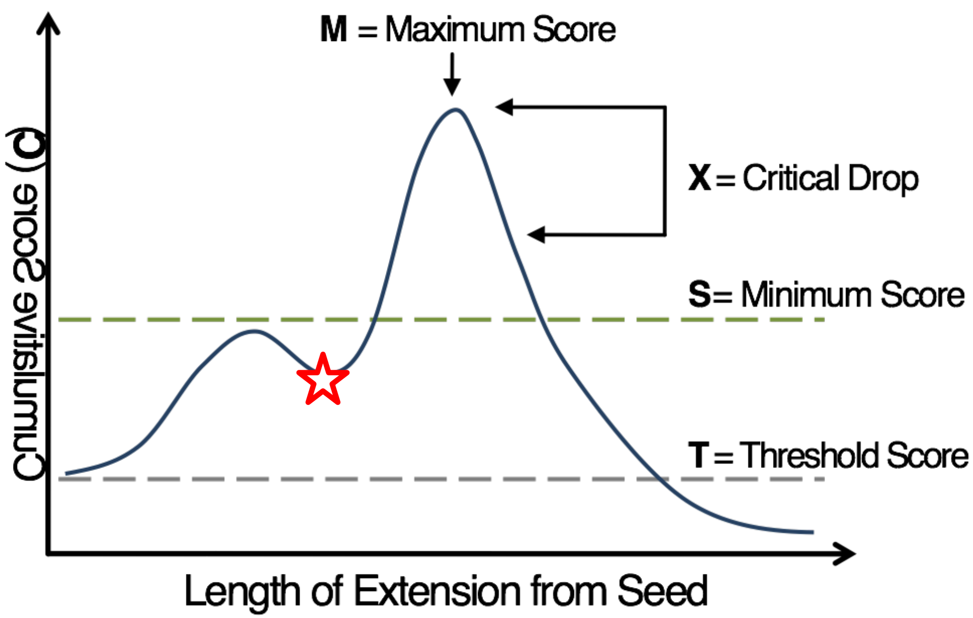
T = 10
S = 20
X = 8

M = 15
C = 15

M	R	L	M	G	**A**	**L**	**E**	**D**	**V**	A	L	N	A	I
C	Q	D	M	C	**S**	**L**	**E**	**E**	**I**	T	L	W	C	Q
					+1	+4	+5	+2	+3					

Extend the seed (both directions!), update **C** and **M**.

Seed Extension



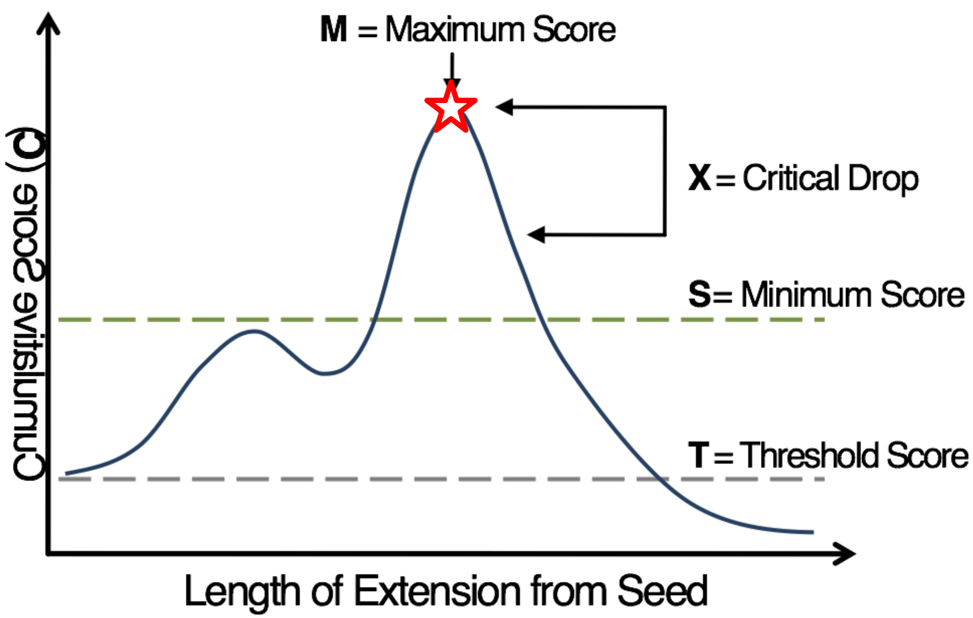
Example:

T = 10
S = 20
X = 8

M = 15
C = 12

M	R	L	M	**G**	**A**	**L**	**E**	**D**	**V**	**A**	L	N	A	I
C	Q	D	M	**C**	**S**	**L**	**E**	**E**	**I**	**T**	L	W	C	Q
				-3	+1	+4	+5	+2	+3	0				

Extend the seed, update **C** but not **M** (we've seen higher M)



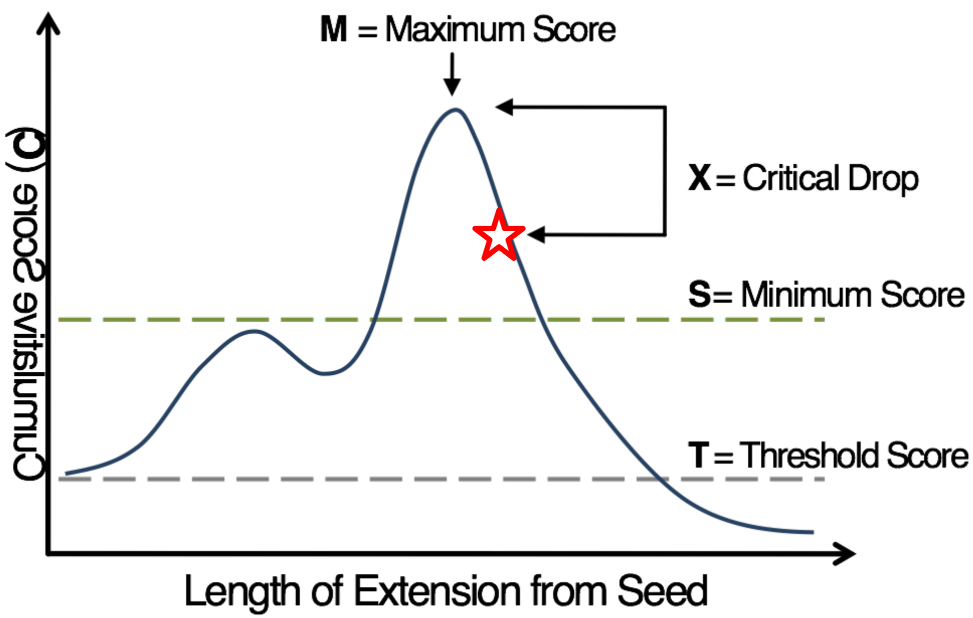
Example:

T = 10
S = 20
X = 8

M = 21
C = 21

M	R	L	**M**	**G**	**A**	**L**	**E**	**D**	**V**	**A**	**L**	N	A	I
C	Q	D	**M**	**C**	**S**	**L**	**E**	**E**	**I**	**T**	**L**	W	C	Q
			+5	−3	+1	+4	+5	+2	+3	0	+4			

Extend the seed, update **C** and **M**.



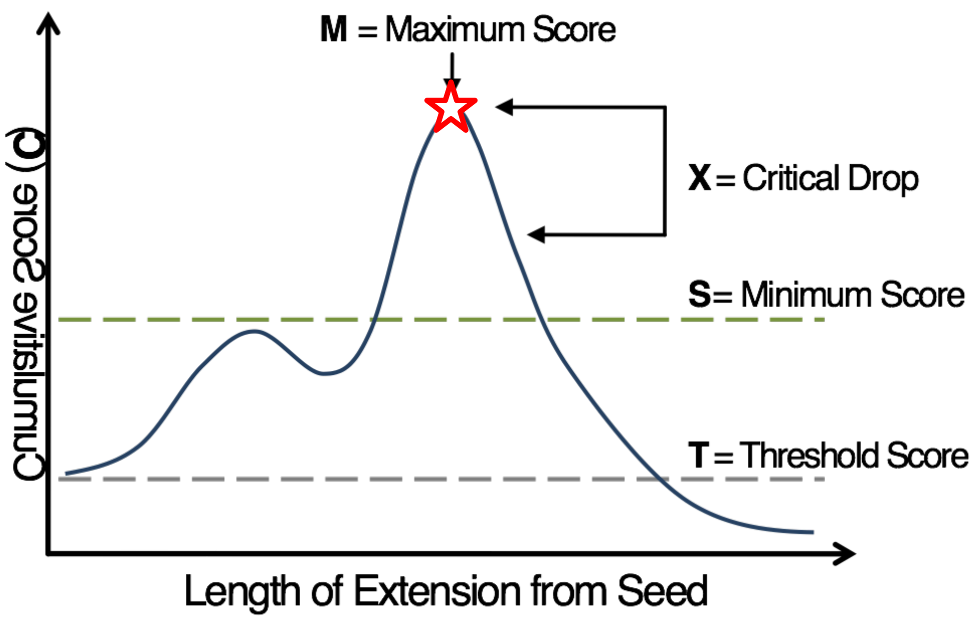
Example:

T = 10
S = 20
X = 8

M = 21
C = 13

M	R	**L**	**M**	**G**	**A**	**L**	**E**	**D**	**V**	**A**	**L**	**N**	A	I
C	Q	**D**	**M**	**C**	**S**	**L**	**E**	**E**	**I**	**T**	**L**	**W**	C	Q
1	−1	−4	+5	−3	+1	+4	+5	+2	+3	0	+4	−4		

Extend the seed, update **C** but not **M**; stop: **C** is **X** less than **M**.



Example:

T = 10
S = 20
X = 8

M = 21
C = 21

M	R	L	**M**	**G**	**A**	**L**	**E**	**D**	**V**	**A**	**L**	N	A	I
C	Q	D	**M**	**C**	**S**	**L**	**E**	**E**	**I**	**T**	**L**	W	C	Q
			+5	−3	+1	+4	+5	+2	+3	0	+4			

M > **S**, so this extension is a high-scoring segment pair.

On the board:
MSA SP Scores

(must answer)
Include all Sums!!
For every Pair!!

In general, show all calculations in the exam notebook

SP Score

How can we score multiple sequence alignments?

We can use a *sum of pairs* score, which is the sum of all pairwise comparisons at a given position of the alignment.

How many pairs?

F	-	I_1	T
F	L	I_2	T
F	R	-	Y
F	R	E	T
+36	−2	−5	9

= 38

Combination!

Binomial coefficient.

$$\binom{4}{2}$$

4 choose 2

4C2

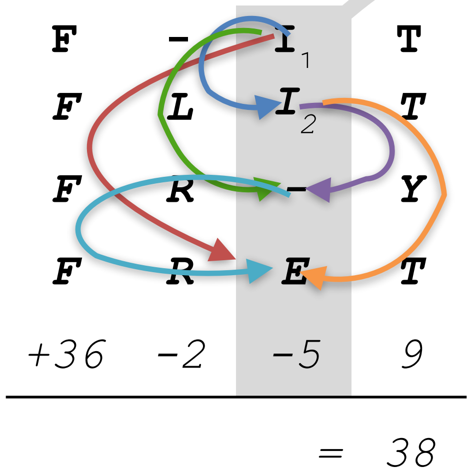
C(n,r) = **n! / r!** (n – r)!

C(4,2) = 4! / 2! (4 – 2)! = 6

SP Score

How can we score multiple sequence alignments?
We can use a *sum of pairs* score, which is the sum of all pairwise comparisons at a given position of the alignment.

6 pairs for a single position in 4 sequences:



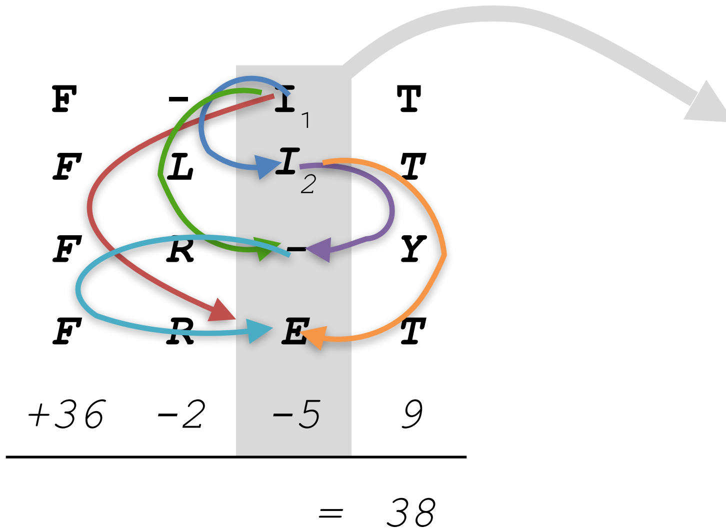
1	2	3	4	5	6
I_1	I_1	I_1	I_2	I_2	-
I_2	-	E	-	E	E
+4	-1	-3	-1	-3	-1
				=	-5

Remember, there are more pairs than sequences.
Go to google and type "4 choose 2" if unsure.

How can we score multiple sequence alignments?

We can use a *sum of pairs* score, which is the sum of all pairwise comparisons at a given position of the alignment.

6 pairs for a single position in 4 sequences:



Pair	First AA	Second AA	BLOSUM score	Sum
1	I1	I2	4	4
2	I1	-	-1	3
3	I1	E	-3	0
4	I2	-	-1	-1
5	I2	E	-3	-4
6	-	E	-1	-5

How can we score multiple sequence alignments?
We can use a *sum of pairs* score, which is the sum of all pairwise comparisons at a given position of the alignment.

6 pairs for a single position in 4 sequences:

F	-	I_1	T
F	L	I_2	T
F	R	-	Y
F	R	E	T
+36	−2	−5	9

= 38

What is the BLOSUM score of F to F ? (Without looking at the matrix)

How can we score multiple sequence alignments?
We can use a *sum of pairs* score, which is the sum of all pairwise comparisons at a given position of the alignment.

6 pairs for a single position in 4 sequences:

F	**-**	**I_1**	**T**
F	***L***	***I_2***	***T***
F	***R***	**-**	***Y***
F	***R***	***E***	***T***
+36	*−2*	*−5*	*9*

= *38*

What is the BLOSUM score of F to F ? *(Without looking at the matrix)*

Answer: 36 / 6 pairs = 6

BLOSUM62 matrix will be provided at the exam (print it out ahead of time)

Ala	A	4																			
Arg	R	-1	5																		
Asn	N	-2	0	6																	
Asp	D	-2	-2	1	6																
Cys	C	0	-3	-3	-3	9															
Gln	Q	-1	1	0	0	-3	5														
Glu	E	-1	0	0	2	-4	2	5													
Gly	G	0	-2	0	-1	-3	-2	-2	6												
His	H	-2	0	1	-1	-3	0	0	-2	8											
Ile	I	-1	-3	-3	-3	-1	-3	-3	-4	-3	4										
Le	L	-1	-2	-3	-4	-1	-2	-3	-4	-3	2	4									
Lys	K	-1	2	0	-1	-3	1	1	-2	-1	-3	-2	5								
Me	M	-1	-1	-2	-3	-1	0	-2	-3	-2	1	2	-1	5							
Ph	F	-2	-3	-3	-3	-2	-3	-3	-3	-1	0	0	-3	0	6						
Pro	P	-1	-2	-2	-1	-3	-1	-1	-2	-2	-3	-3	-1	-2	-4	7					
Ser	S	1	-1	1	0	-1	0	0	0	-1	-2	-2	0	-1	-2	-1	4				
Thr	T	0	-1	0	-1	-1	-1	-1	-2	-2	-1	-1	-1	-1	-2	-1	1	5			
Trp	W	-3	-3	-4	-4	-2	-2	-3	-2	-2	-3	-2	-3	-1	1	-4	-3	-2	11		
Tyr	Y	-2	-2	-2	-3	-2	-1	-2	-3	2	-1	-1	-2	-1	3	-3	-2	-2	2	7	
Val	V	0	-3	-3	-3	-1	-2	-2	-3	-3	3	1	-2	1	-1	-2	-2	0	-3	-1	4
		A	R	N	D	C	Q	E	G	H	I	L	K	M	F	P	S	T	W	Y	V
		Ala	Arg	Asn	Asp	Cys	Gln	Glu	Gly	His	Ile	Le	Lys	Me	Ph	Pro	Ser	Thr	Trp	Tyr	Val

Sequence similarity BLAST STATISTICS

Ala	A	4																			
Arg	R	-1	5																		
Asn	N	-2	0	6																	
Asp	D	-2	-2	1	6																
Cys	C	0	-3	-3	-3	9															
Gln	Q	-1	1	0	0	-3	5														
Glu	E	-1	0	0	2	-4	2	5													
Gly	G	0	-2	0	-1	-3	-2	-2	6												
His	H	-2	0	1	-1	-3	0	0	-2	8											
Ile	I	-1	-3	-3	-3	-1	-3	-3	-4	-3	4										
Leu	L	-1	-2	-3	-4	-1	-2	-3	-4	-3	2	4									
Lys	K	-1	2	0	-1	-3	1	1	-2	-1	-3	-2	5								
Met	M	-1	-1	-2	-3	-1	0	-2	-3	-2	1	2	-1	5							
Phe	F	-2	-3	-3	-3	-2	-3	-3	-3	-1	0	0	-3	0	6						
Pro	P	-1	-2	-2	-1	-3	-1	-1	-2	-2	-3	-3	-1	-2	-4	7					
Ser	S	1	-1	1	0	-1	0	0	0	-1	-2	-2	0	-1	-2	-1	4				
Thr	T	0	-1	0	-1	-1	-1	-1	-2	-2	-1	-1	-1	-1	-2	-1	1	5			
Trp	W	-3	-3	-4	-4	-2	-2	-3	-2	-2	-3	-2	-3	-1	1	-4	-3	-2	11		
Tyr	Y	-2	-2	-2	-3	-2	-1	-2	-3	2	-1	-1	-2	-1	3	-3	-2	-2	2	7	
Val	V	0	-3	-3	-3	-1	-2	-2	-3	-3	3	1	-2	1	-1	-2	-2	0	-3	-1	4
		A	R	N	D	C	Q	E	G	H	I	L	K	M	F	P	S	T	W	Y	V
		Ala	Arg	Asn	Asp	Cys	Gln	Glu	Gly	His	Ile	Leu	Lys	Met	Phe	Pro	Ser	Thr	Trp	Tyr	Val